# Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

Auckland Council
Te Kaunihera o Tāmaki Makaurau

## PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

| | |
|---|---|
| Cover sheet issued on behalf of: | Royal Glass |
| In respect of (system type): | Opus Vista System |
| Supplied to: | Auckland Council |
| Client / designers name: | Josh Harrison |
| Description of work: *E.g. New dwelling* | 12mm thick Glass Balustrade installation for External Balcony area using Channel System |
| Address of property: | 83 Speights Road, St Heiliers, Auckland 1071 |

## PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

| | | | | | |
|---|---|---|---|---|---|
| Deck / balcony / stairs construction: | ☑ Timber | ☐ Precast | ☐ Concrete | ☐ Steel | ☐ *please specify* |
| Location of deck / balcony / stairs: | ☐ Internal | ☑ External | ☐ *please specify* | | |
| Deck / balcony / stairs: | ☑ New | ☐ Existing | Cladding: | ☑ Direct fixed | ☐ Cavity ☐ N/A |

## PART 3: GLASS DETAILS *(tick boxes that apply)*

| | | | |
|---|---|---|---|
| Type: | ☑ Toughened | ☑ Toughened laminated | ☐ Toughened laminated with stiff interlayer |
| Thickness: | 12mm (mm) | Width of pane: | 0.5-1.8 (m) |
| Height above floor: | (m) | Height above fixing: | (m) |

**Note:** If barrier extends more than 1.2m above fixing point, test reports must be provided

| | |
|---|---|
| Supplier: | Royal Glass |

## PART 4: BARRIER DETAILS

| | |
|---|---|
| System name: | Opus Vista System |
| Drawings (ref page number) | |
| Engineers name: | |
| Hardware (type): | |

Drawings <u>must</u> show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

## PART 5: SAFETY FEATURES *(tick all boxes that apply)*

- ☑ Continuous interlinking rail (show how fixed to building and each pane of glass)
- ☐ Continuous interlinking top cap (show how fixed to building and each pane of glass)
- ☐ Clamps, brackets or point fixings (show how fixed to building and each pane of glass)
- ☐ Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass)
- ☐ Glass with stiff interlayer
- ☐ Two or three glass edges supported by continuous clamps
- ☐ Structural post to fix interlinking rail to deck
- ☐ Other (specify)

| | |
|---|---|
| Engineers name: | PNP Engineering |
| Drawings (ref page number) | |
| Comments: | |

## PART 6: COVER SHEET COMPLETED BY

| | |
|---|---|
| Name: | Roxy Huang |
| Signature: | |
| Date: | 14/05/2025 |
| Contact number: | 0800 769 254 |
| Email: | info@royalglass.co.nz |

## FOR RESIDENTIAL OCCUPANCY A, B, C3 & E OF TABLE 3.3 AS/NZS 1170.1

**NOTES:**

1) This proprietary balustrade system complies with New Zealand Building Code Clause B1 Structure, B1/AS1 Amendment 15, B1/VM1, F2 Hazardous Building Materials and F4 Safety From Falling Third Addition, subject to:
- all products meeting the required performance specification
- site installation carried out in accordance with intent of this drawing and comply with NZS 4223.3.2016 requirements

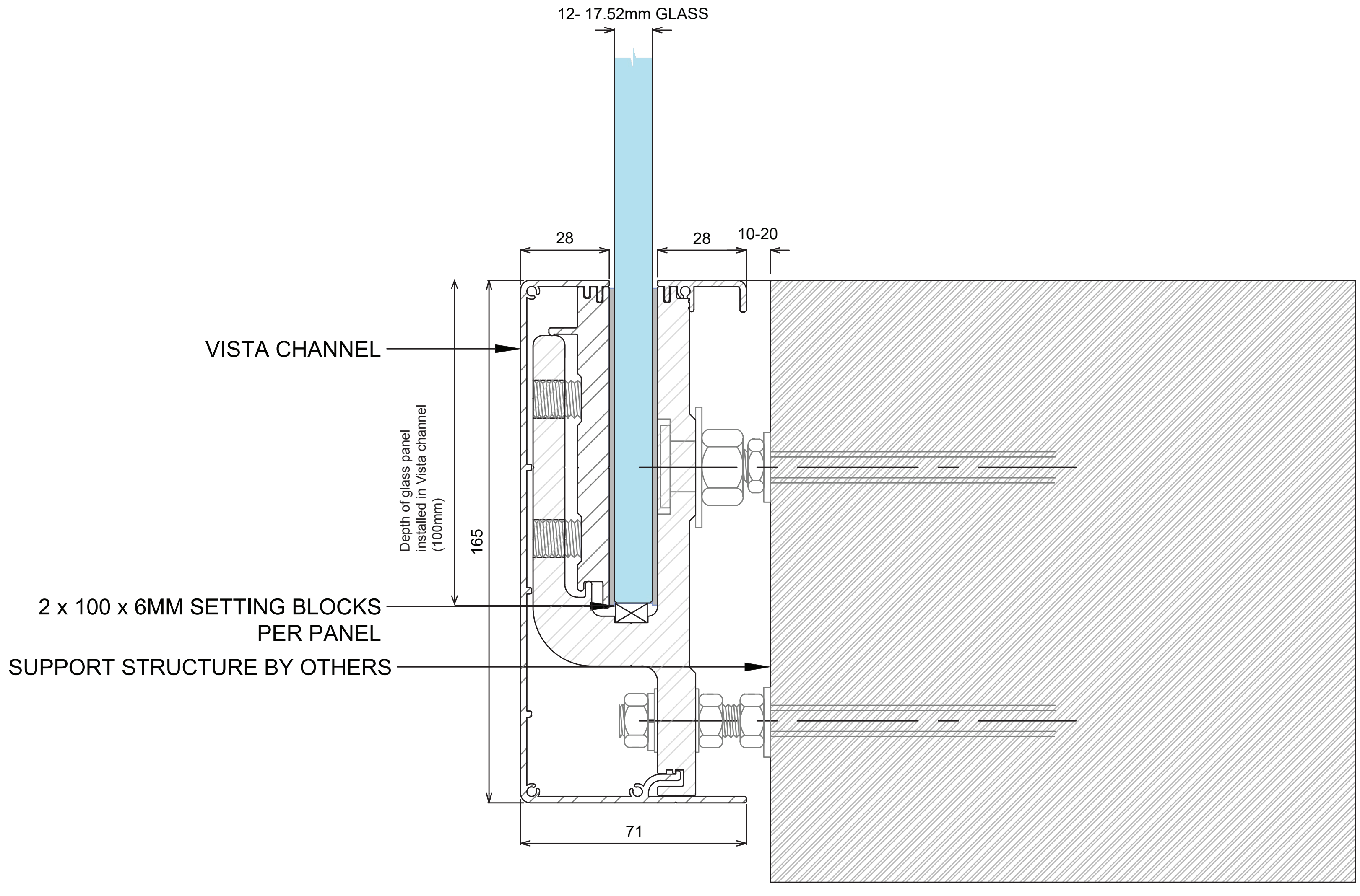
2) Heights are measured from the base at the top of the clamp to the top of the glass

3) Design of support structure is the responsibility of others

4) A handrail of 32-50mm diameter is required for stairs and ramps exceeding 1:20 slope. Refer NZBC D1/AS1

5) Safety glass options according to 22.4.3 of NZS 4223.3:2016 are:

a. 12mm Toughened Satety Glass with Interlinking rail
b. 13.2mm Toughened Laminated Glass with glass gap clamps and a maximum pane width of 2000mm
c. 13.52mm Toughened SentryGlas Laminated Glass with a minimum pane width of 1000mm
d. 15mm Toughened Safety Glass with interlinking rail
e. 17.52mm Toughened SentryGlas Laminated Glass with minimum pane width of 1000mm



## Side Fix VISTA - Safety From Falling Barriers:

Concrete, steel & timber substrates only.
LAMINATED SENTRY toughened laminated safety glass, and TOUGHENED monolithic toughened safety glass.

| Glass Thickness t (mm) | Occupancy | Maximum Design Height H (mm) | Channel Fixing Spacing (mm) Max | Design loads to deck structure | | Maximum design wind pressure | |
|---|---|---|---|---|---|---|---|
| | | | | M* (kNm/spacing) | T* (kN) | SLS Wind (kPa) | ULS Wind (kPa) |
| 12 | A Internal | 1100 (Concrete & Steel) | 600 | TBC | TBC | | |
| | | 1100 (Timber) | 300 | TBC | TBC | | |
| 12 | A, B, C3 & E | 1100 (Concrete & Steel) | 600 | 0.93 | 10.32 | 1.67 (Extra High) | 2.13 (Extra High) |
| | | 1100 (Timber) | 300 | 0.46 | 5.16 | 1.67 (Extra High) | 2.13 (Extra High) |
| 13.52 | A, B, C3 & E | 1100 (Concrete & Steel) | 600 | 0.93 | 10.32 | 1.67 (Extra High) | 2.13 (Extra High) |
| | | 1100 (Timber) | 300 | 0.46 | 5.16 | 1.67 (Extra High) | 2.13 (Extra High) |
| 15 | A Internal | 1250 (Concrete & Steel) | 600 | TBC | TBC | | |
| | | 1250 (Timber) | 300 | TBC | TBC | | |
| 15 | A, B, C3 & E | 1250 (Concrete & Steel) | 600 | TBC | TBC | 1.67 (Extra High) | 2.13 (Extra High) |
| | | 1200 (Timber) | 300 | TBC | TBC | 1.67 (Extra High) | 2.13 (Extra High) |
| 17.52 | A, B, C3 & E | 1250 (Concrete & Steel) | 600 | TBC | TBC | 1.67 (Extra High) | 2.13 (Extra High) |
| | | 1200 (Timber) | 300 | TBC | TBC | 1.67 (Extra High) | 2.13 (Extra High) |

T* is tension force per fastener
(@ 200 or 400mm centres)

## Side Fix VISTA - Pool Fence (not protecting a fall > 1m):

Concrete, steel & timber only

| Glass Thickness t (mm) | NZS 3604 Wind Zone | Substrate | Maximum Design Height H (mm) | Channel Fixing Spacing (mm) Max | Design loads to deck structure | |
|---|---|---|---|---|---|---|
| | | | | | M* (kNm/spacing) | T* (kN) |
| 12 | Medium | Timber | 1300 | 300 | TBC | TBC |
| | | Concrete & Steel | 1300 | 600 | TBC | TBC |
| | High | Timber | 1250 | 300 | TBC | TBC |
| | | Concrete & Steel | 1250 | 600 | TBC | TBC |
| | Very High | Timber | 1200 | 300 | TBC | TBC |
| | | Concrete & Steel | 1250 | 600 | TBC | TBC |
| 15 | Extra High | Timber | 1250 | 300 | TBC | TBC |
| | | Concrete & Steel | 1250 | 600 | TBC | TBC |

## Glass thickness key:

Note: Inner layer refers to balcony / walking surface side

| Glass Thickness t (mm) | Inner layer glass thickness (mm) | Interlayer thickness (mm) and type | Outer layer glass thickness (mm) | Panel size requirements | |
|---|---|---|---|---|---|
| | | | | Minimum panel width (mm) | Maximum panel width (mm) |
| 12 | - | - | - | 1000 | Refer manufacturing limits |
| 13.2 | 6 | 1.2 LAMINATE | 6 | 1000 | Refer manufacturing limits |
| 13.52 | 6 | 1.52 SENTRYGLAS | 6 | 1000 | Refer manufacturing limits |
| 15 | - | - | - | 1000 | Refer manufacturing limits |
| 17.52 | 8 | 1.52 SENTRYGLAS | 8 | 1000 | Refer manufacturing limits |

## Maximum panel widths for Interlinking Rail/Bracket systems:

Applies where barrier is protecting a fall of 1.0m or more

| Interlinking Rail System | Maximum panel width (mm) | Position |
|---|---|---|
| S25 | 1700 | on glass only |
| S40 | 1700/1900 | HB50 bracket / on glass |
| R40 | 1700/1900 | HB50 bracket / on glass |
| AH40 | 1700/1900 | HB50 bracket / on glass |

## Post failure requirements:

Applies where barrier is protecting a fall of 1.0m or more

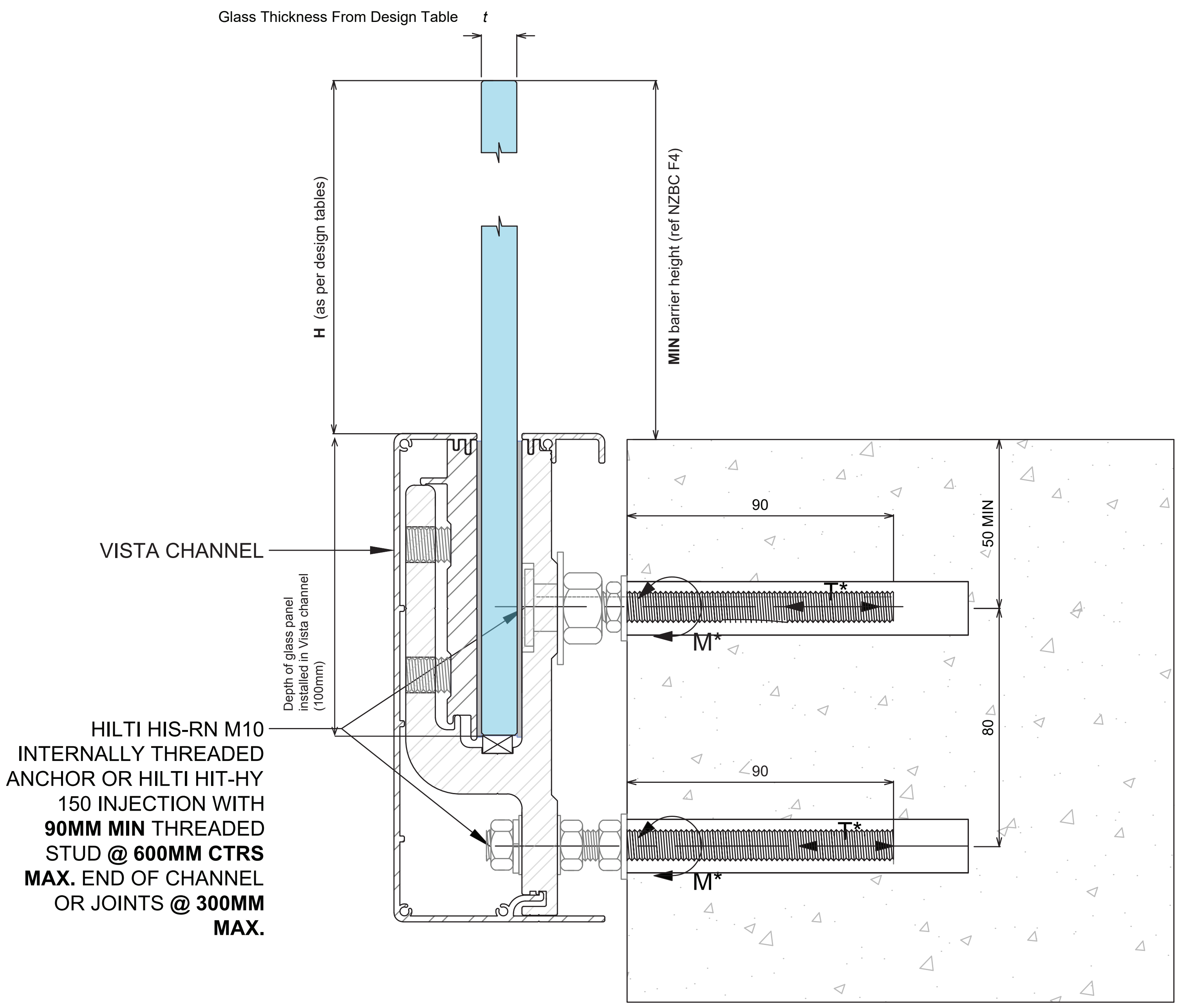
| | |
|---|---|
| TOUGHENED | Interlinking rail required in all cases |
| LAMINATED STF | No interlinking rail required<br>Minimum panel widths apply |
| SENTRY | No interlinking rail required<br>Minimum panel widths apply |

**NOTES:**
1) Design table only valid for use with VISTA balustrade system.
2) Refer to elevation drawings for Height 'H'.
3) The specifier must ensure the balustrade height above floor level requirements as per the NZ Building Code are complied with.
4) Design loads are in accordance with AS/NZS 1170.1:2002 table 3.3 as modified by NZBC B1/VM1 and DBH Guidance on Barrier Design (March 2012).
5) M* & T* denote bending moment (kNm/m width) and tension loads (kN/connection) respectively to be supported by the deck/pool structure.
6) T* is based on wet SG8 timber substructures (except for occupancy A).
7) Capacity of all substructure is to be verified by building engineer or checked for accordance with NZS3604 (where applicable) prior to fixing.
8) All glass is to be toughened safety, in either TOUGHENED Monolithic or LAMINATED SENTRY variants subject to requirements of the tables above. Minimum glass residual strength 100MPa, all edges polished.
9) Glass & interlayer thicknesses shown are nominal thickness. Table is based on glass minimum tolerance as per NZS 4223.1:2008.
10) Refer to the relevant fixing details on drawings
11) The tables for this balustrade system are based on an SLS deflection limit of 50mm. While greater than the suggested limit of height/60 as specified in NZS1170.0 for post and rail handrail systems, this is deemed acceptable based on the nature of the cantilevered glass system.
12) For designs outside the scope of these tables and ULS wind pressures exceeding those shown, specific design is required.
13) Maximum 10mm tolerance allowed to H heights noted in table.
14) *Very High wind zone requires a structural nib wall, fit to fence inside face.
15) LUGANO front cover does not posess toe hold.

**FOR RESIDENTIAL OCCUPANCY A, B, C3 & E OF TABLE 3.3 AS/NZS 1170.1**

## NOTES:

1) This proprietary balustrade system complies with New Zealand Building Code Clause B1 Structure, B1/AS1 Amendment 15, B1/VM1, F2 Hazardous Building Materials and F4 Safety From Falling Third Addition, subject to:
- all products meeting the required performance specification
- site installation carried out in accordance with intent of this drawing




## NOTES:

1) Capacity of structure is to be of sufficient strength to support loads M* and T* specified on balustrade system design table. Structure capacity to be verified by building engineer prior to fixing balustrade.
2) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with balustrade system design table.
3) No substitution allowed - any variation from the details above and design tables will require specific design.
4) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

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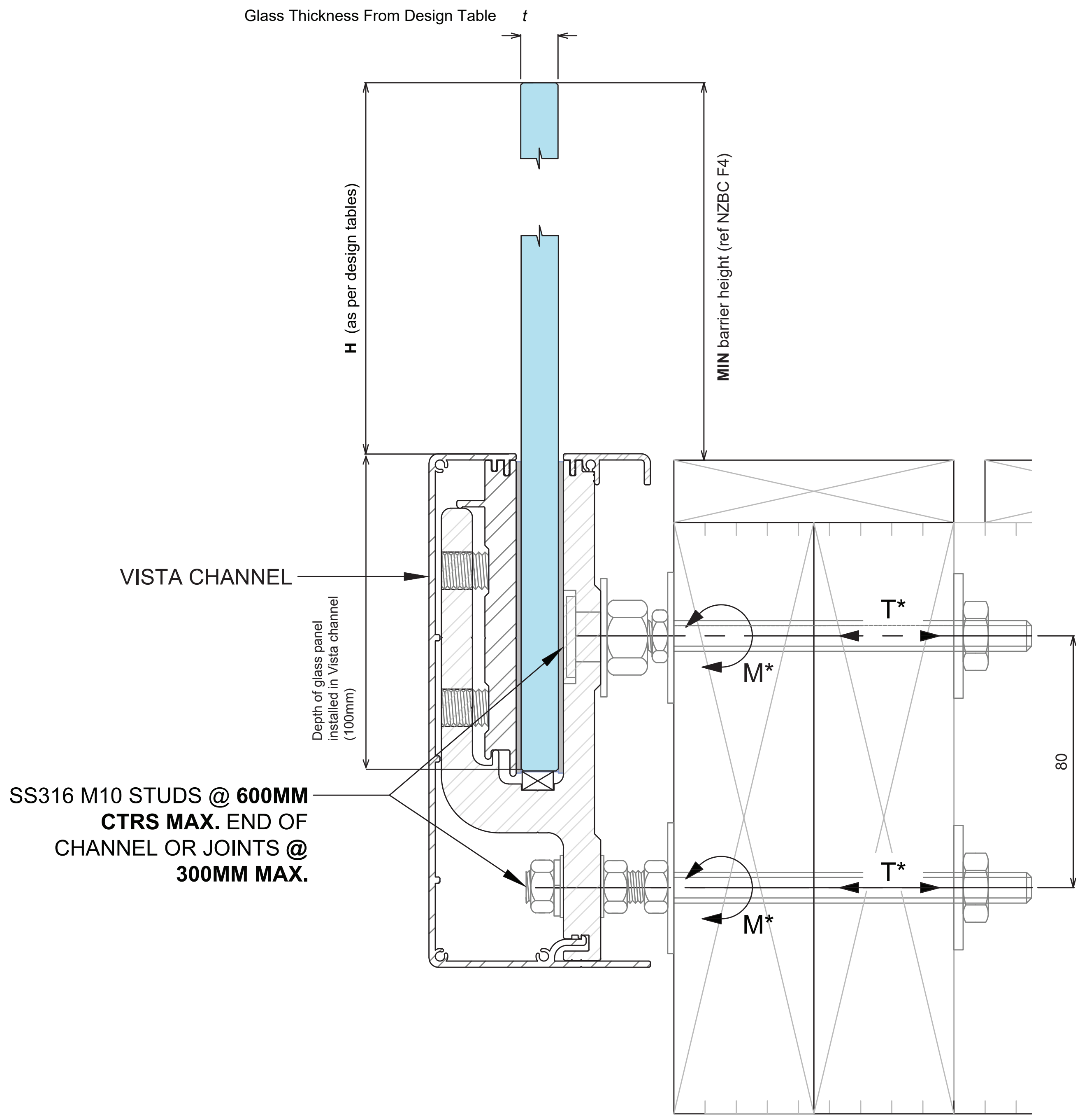


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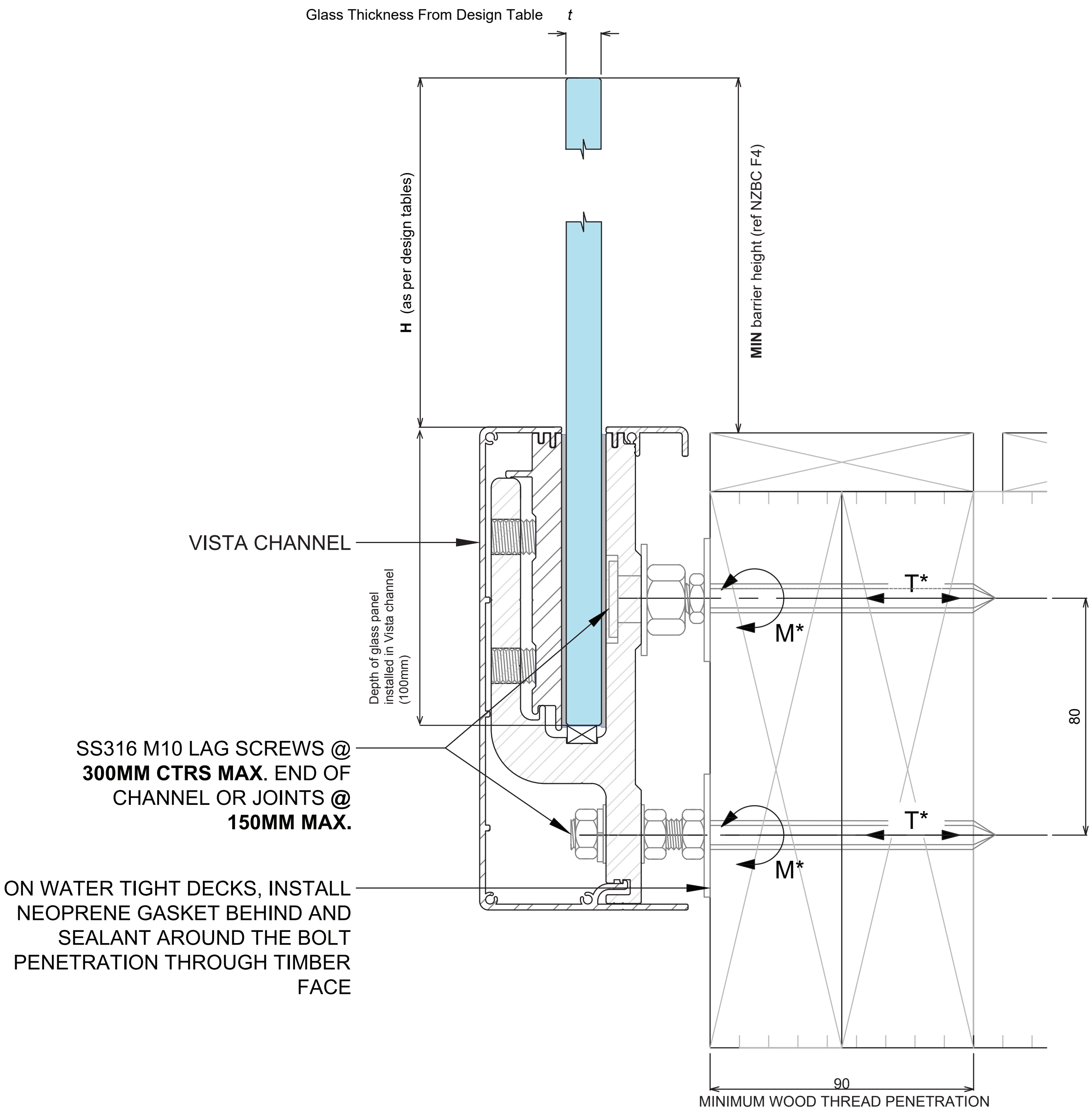


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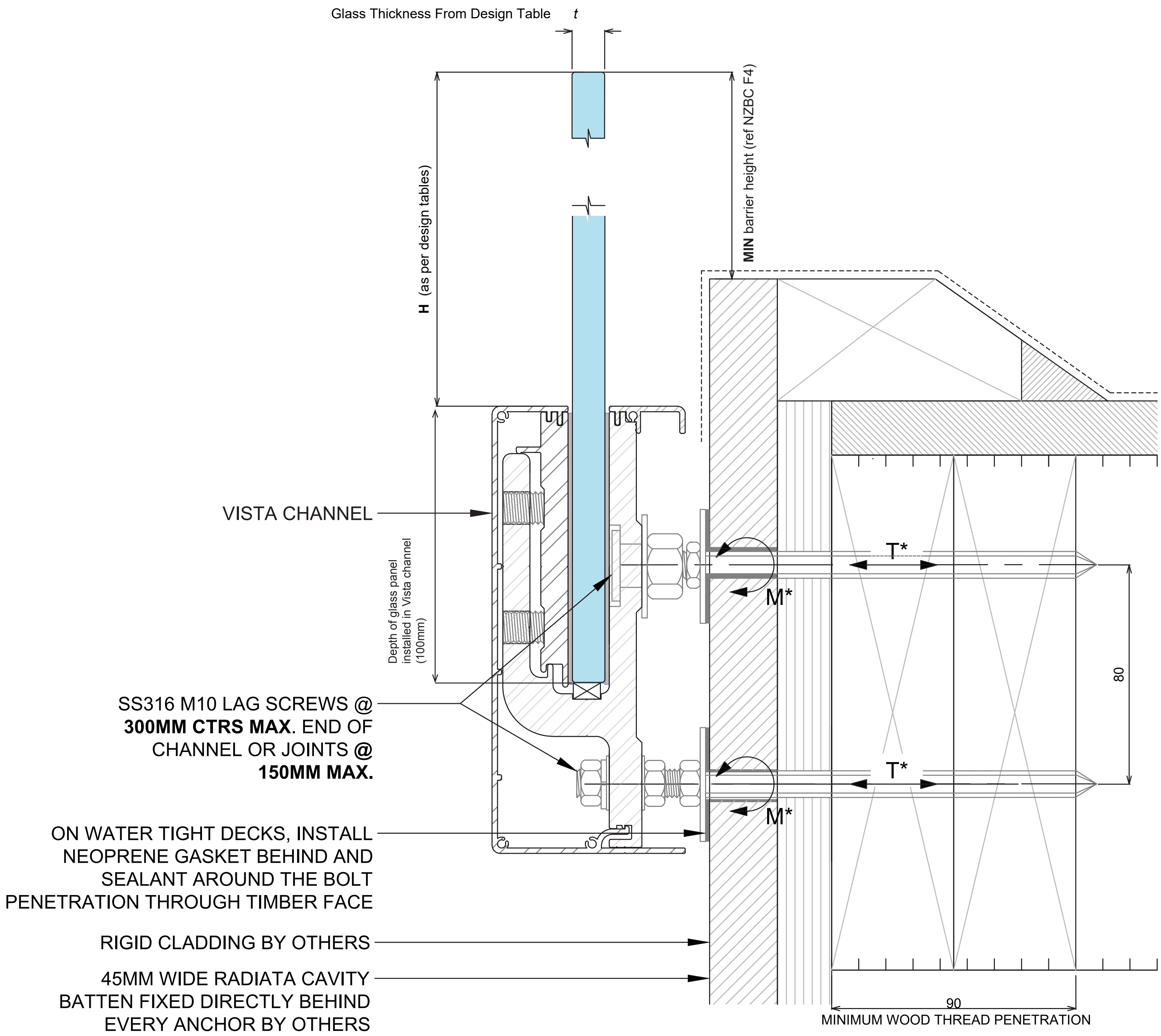


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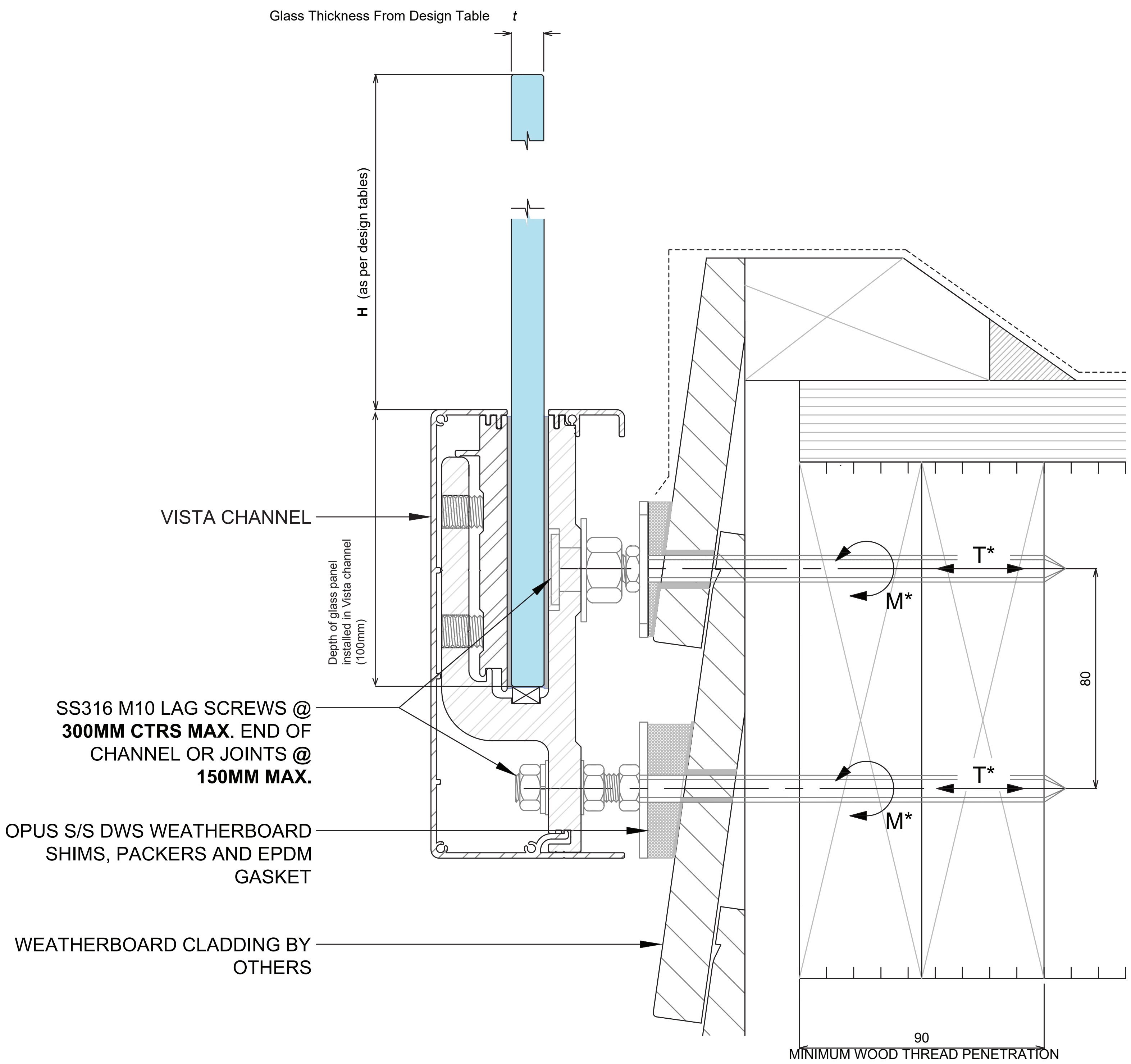


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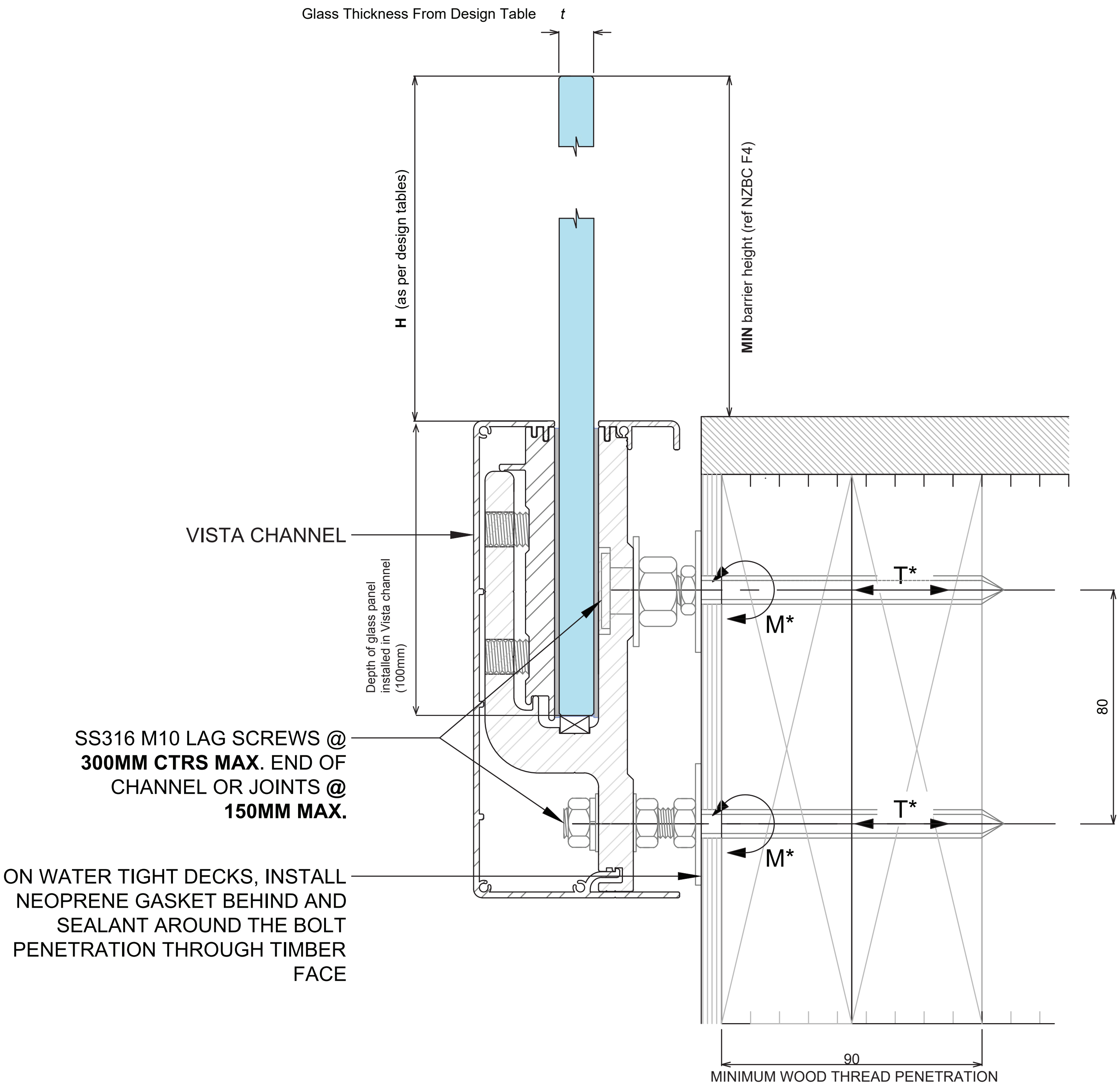


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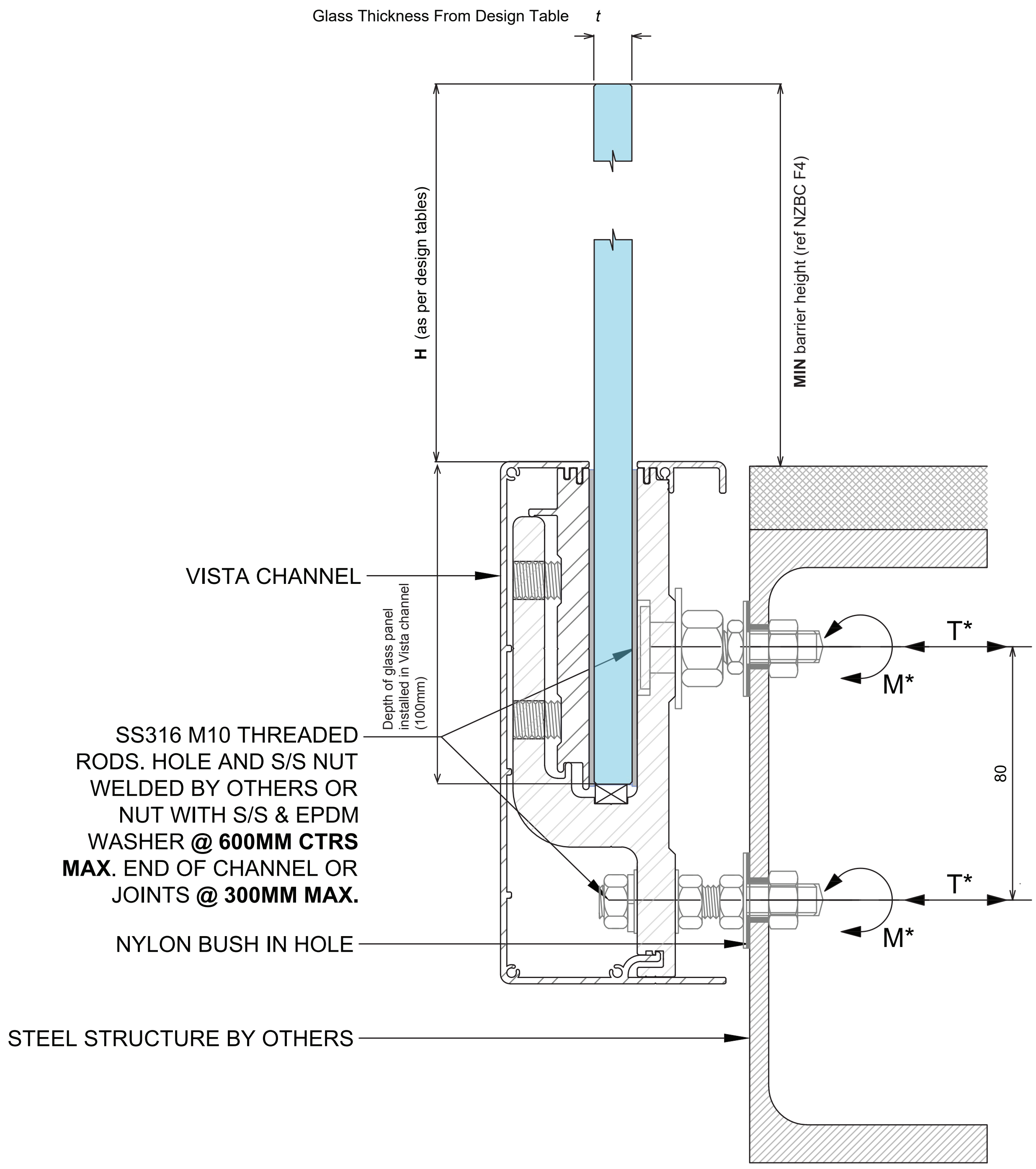


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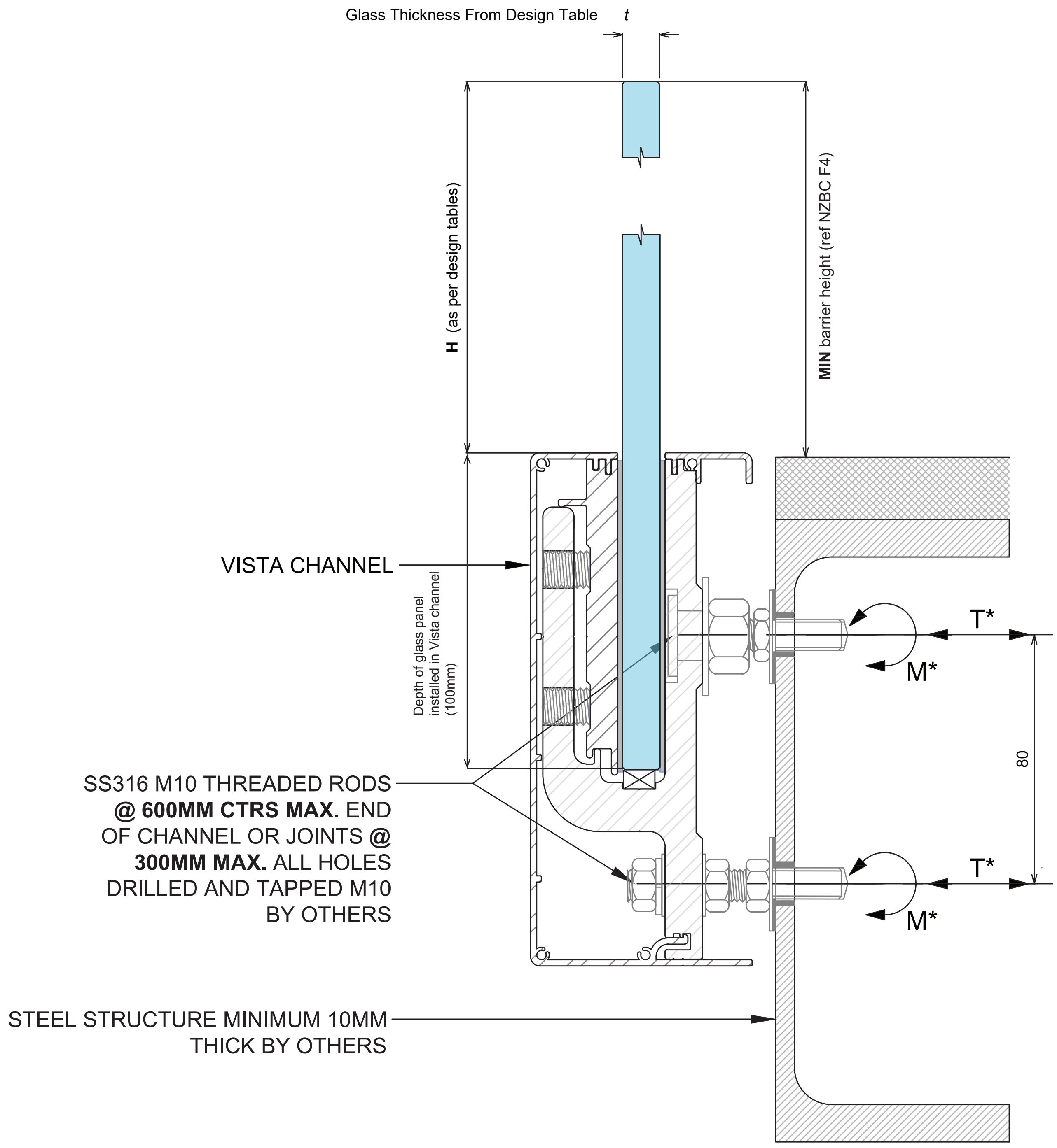


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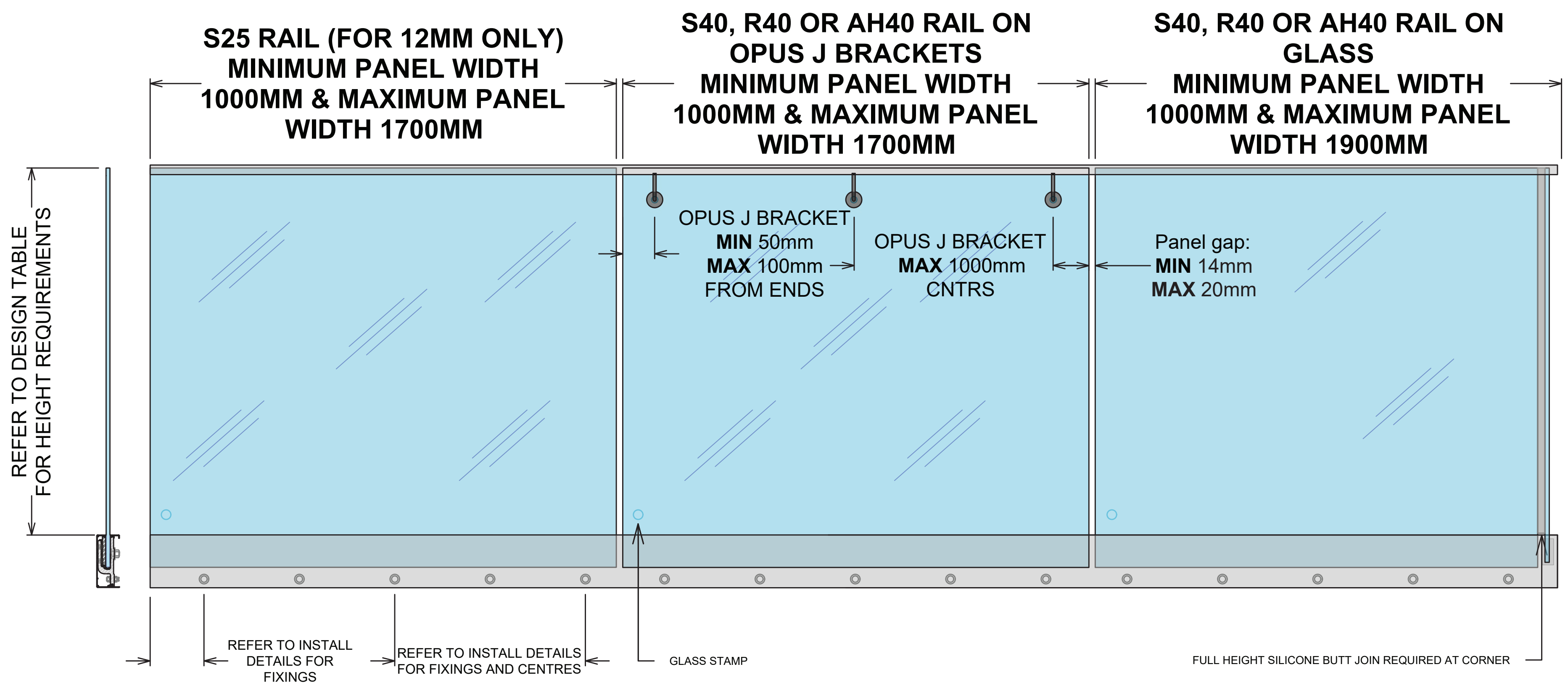
## 12mm & 15mm TOUGHENED SAFETY GLASS

**PANEL WIDTH NOTES:**
Minimum panel width where two or more panels are in a straight line = 1000mm.
Minimum width for short return panel = 200mm.

**GLASS & FIXING SPECIFICATIONS**:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

**INTERLINKING RAIL REQUIRED:**
S25 (for 12mm glass on glass only, **MAX** 1700mm panels)
S40, R40 OR AH40 (on J brackets, **MAX** 1700mm panels)
S40, R40 OR AH40 (on glass, **MAX** 1900mm panels)
RT50/ ST50 (on handrail brackets only, **MAX** 1700mm panels)



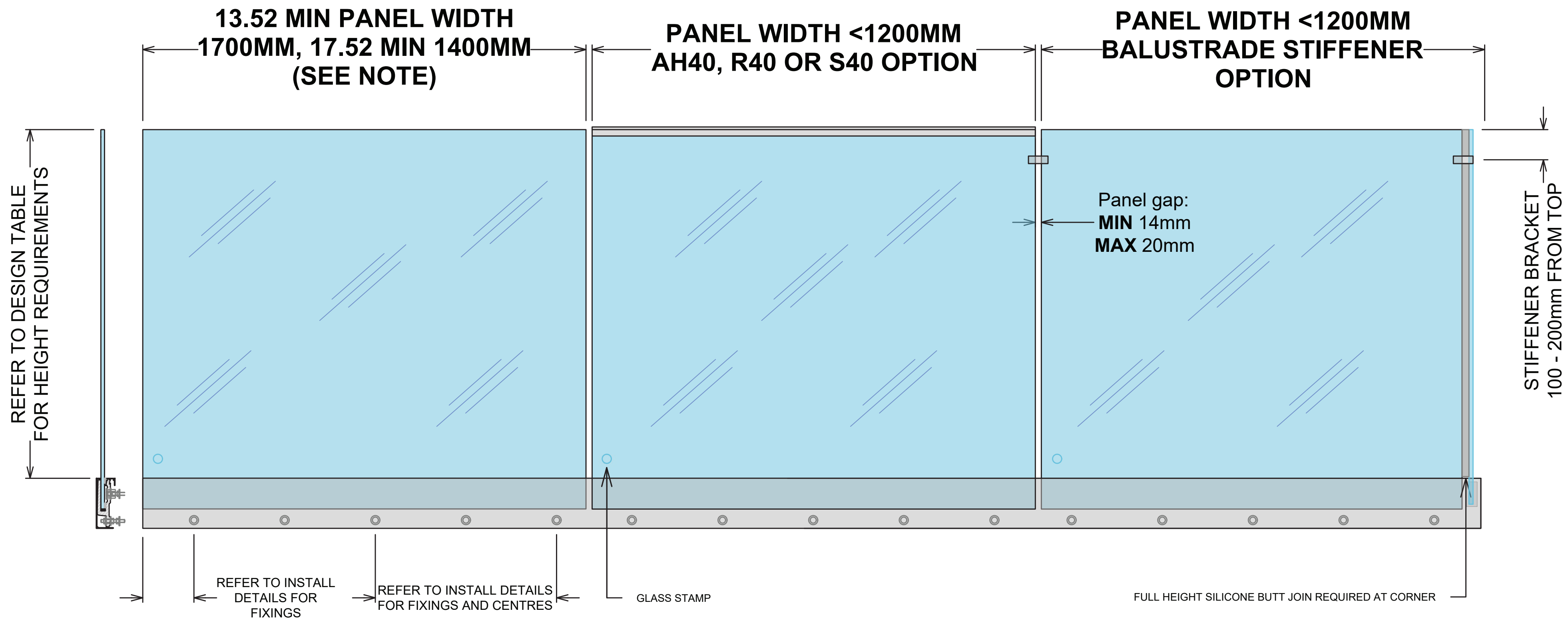

## 13.2mm TOUGHENED LAMINATE SAFETY GLASS, 13.52 & 17.52mm TOUGHENED SENTRYGLAS LAMINATED GLASS

**PANEL WIDTH NOTES:**
Balustrade stiffener brackets or interlinking rail required for panels <1200mm
Minimum panel width where two or more panels are in a straight line = 1000mm
Minimum width for short return panel = 200mm.

**NOTE:**
13.52mm & 17.52mm can be fully frameless
13.2mm must at least have stiffener brackets

**RAIL & STIFFENERS REQUIRED:**
S40, R40 OR AH40 (on J brackets, **MAX** 1700mm panels)
S40, R40 OR AH40 (on glass, **MAX** 1900mm panels)
RT50/ ST50 (on handrail brackets only, **MAX** 1700mm panels)

**GLASS & FIXING SPECIFICATIONS**:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

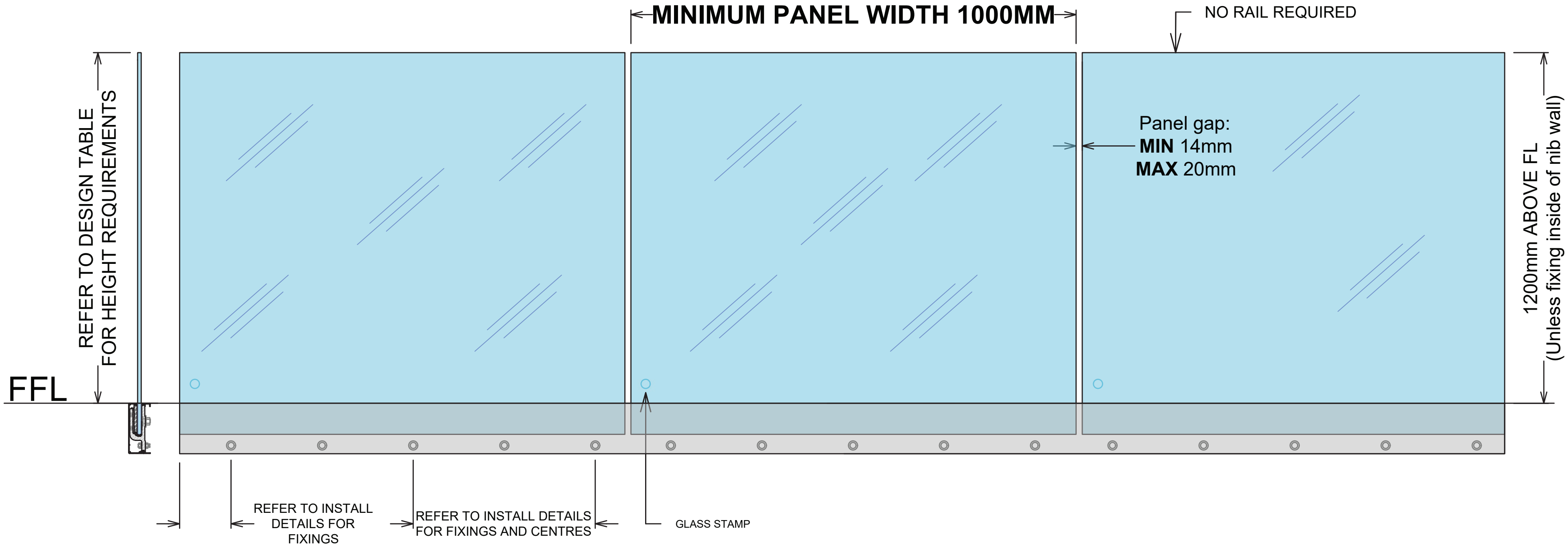


## POOL BALUSTRADE

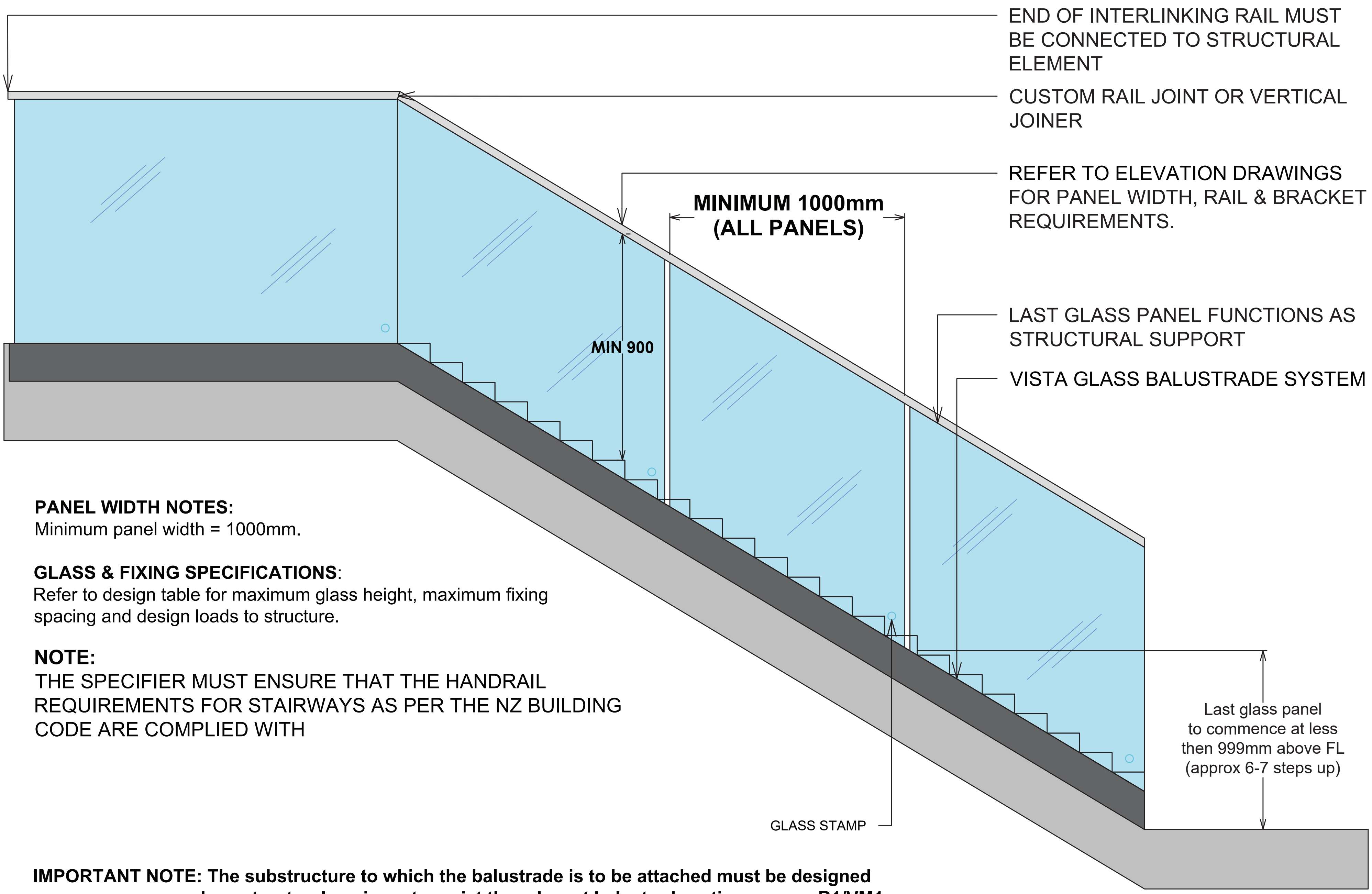
**GLASS & FIXING SPECIFICATIONS**:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

**IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.**

**NOTES:**
APPLIES TO FREE STANDING POOL FENCES NOT PROTECTING A FALL OF >1000mm
AS OF JAN 2017, COMLIES WITH BUILDING CODE CLAUSE F9 & SECTION 162C OF THE BUILDING ACT




## STAIR BALUSTRADE



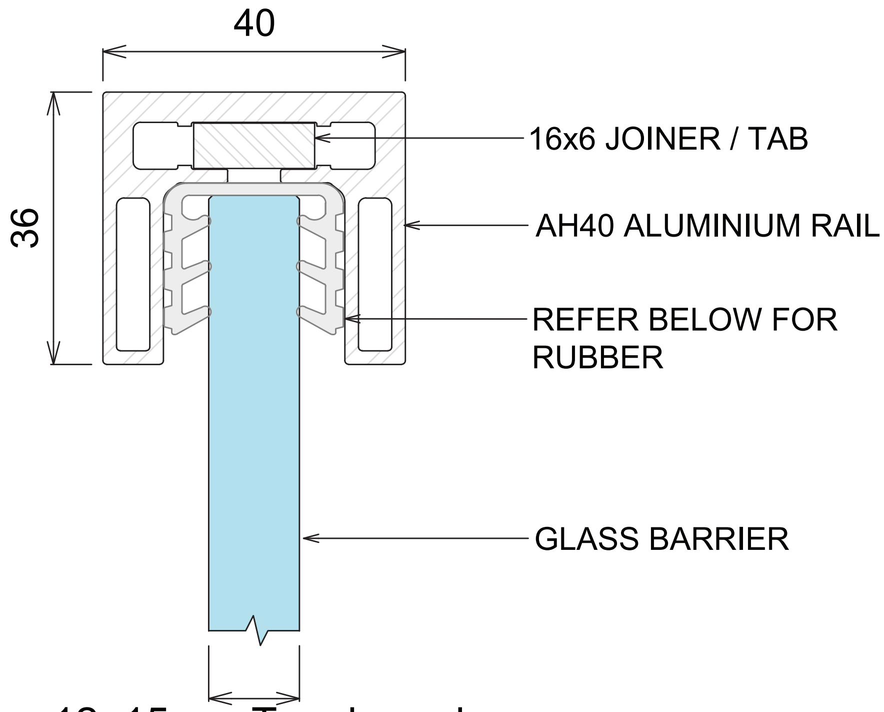

**PANEL WIDTH NOTES:**
Minimum panel width = 1000mm.

**GLASS & FIXING SPECIFICATIONS**:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

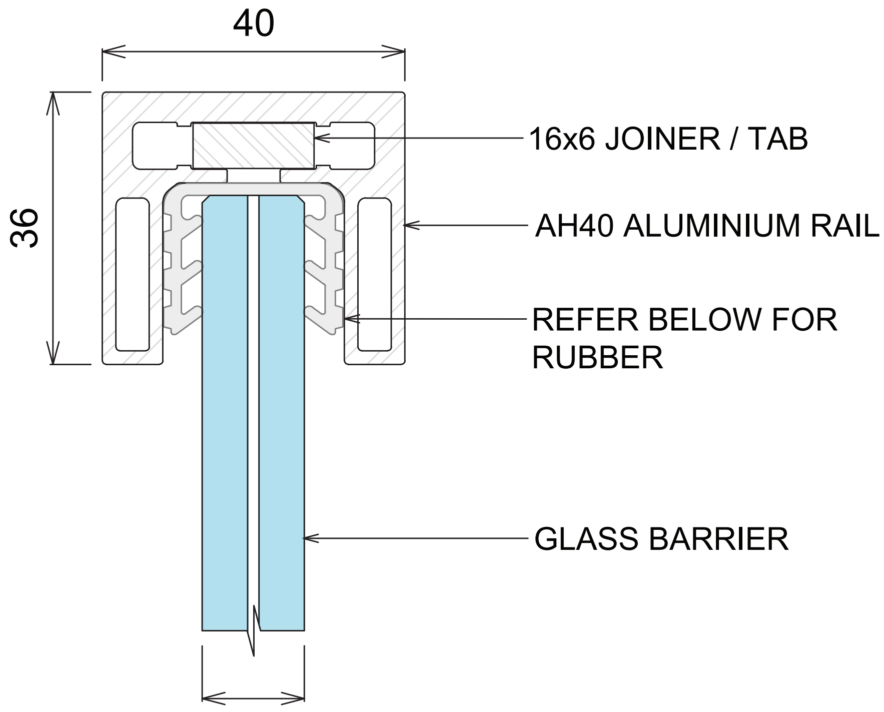
**NOTE:**
THE SPECIFIER MUST ENSURE THAT THE HANDRAIL REQUIREMENTS FOR STAIRWAYS AS PER THE NZ BUILDING CODE ARE COMPLIED WITH

**IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.**

## REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS

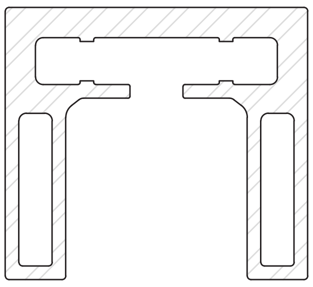



12, 15mm Toughened Safety Glass

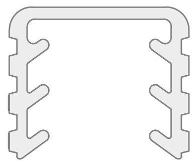



13.2, 13.52, 17.52 Toughened Laminate Safety Glass

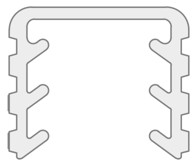
## COMPONENTS



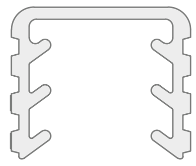
AH40 ALUMINIUM RAIL



AHGR1
FOR 12mm GLASS

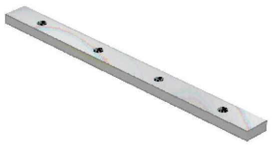


AHGR2
FOR 15mm GLASS



AHGR3
FOR 19mm GLASS

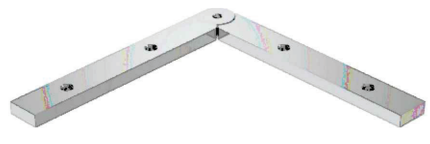
**JOINERS ALL 16 x 6mm**



AHJ180 STRAIGHT INLINE JOINER WITH LOCKING SCREWS

AHJ90 90 DEGREE JOINER WITH LOCKING SCREWS

AHAVJ ADJUSTABLE VERTICAL JOINER WITH LOCKING SCREWS



AHAHJ ADJUSTABLE HORIZONTAL JOINER WITH LOCKING SCREWS

**END CAP**

SATIN BRUSHED ANO AHWB-SS

BRUSHED BLACK ANO AHWB-BK

**WALL BRACKET**

SATIN BRUSHED ANO AHEC-SS

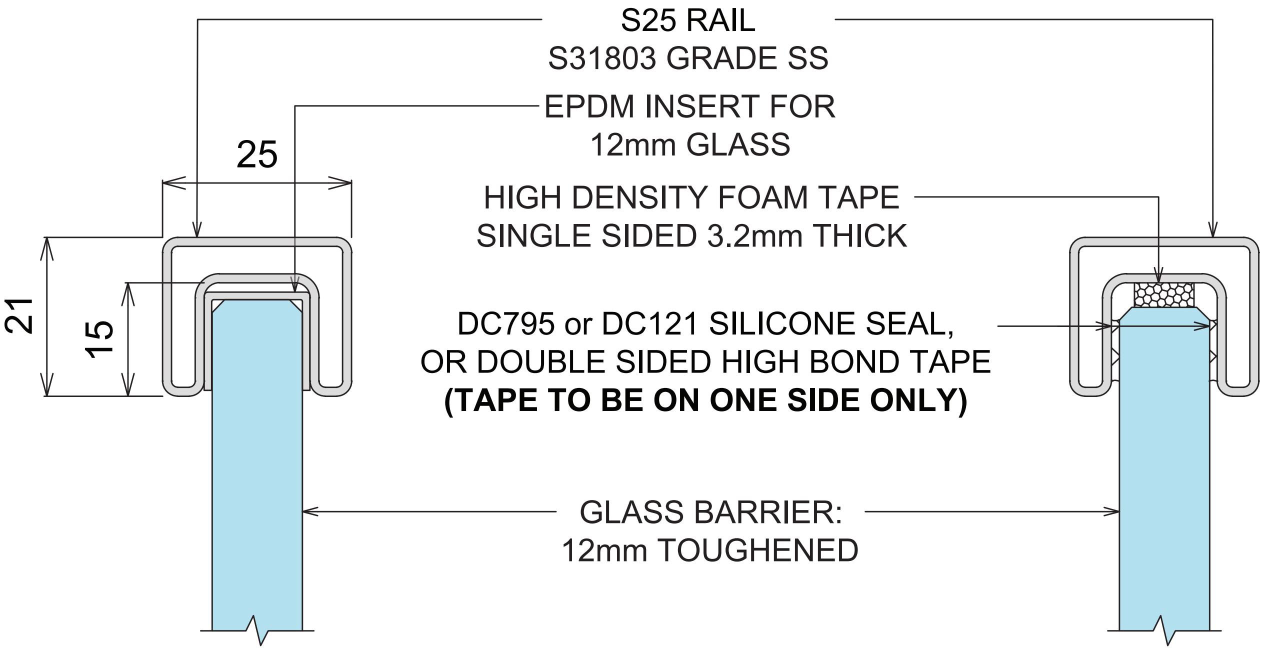
BRUSHED BLACK ANO AHEC-BK

**INSTALLATION NOTES:**

1. CUT SHORT LENGTHS OF GASKET (nom 50mm) AND PLACE AT APPROXIMATELY 700mm CENTRES.
2. CUT / ADJUST INTERLINKING RAIL TO CORRECT DIMENSIONS AND TEST IN POSITION.
3. REMOVE ALL PARTS FROM GLASS BARRIER AND INSTALL FULL CUT LENGTHS OF GASKET TO TOP EDGE OF GLASS BARRIER.
4. ASSEMBLE TOP RAIL, JOINERS AND SUITABLE END PLATES.
5. PLACE BLOBS OF SEIGALSEAL N05+ OR V60 SILICONE IN EVERY GASKET HOLE.
6. PLACE TOP RAIL EXTRUSION, JOINERS AND END PLATES IN POSITION ON GLASS BARRIER, CLIPPING FIRMLY TO GASKET.
7. TAPE ASSEMBLED COMPONENTS DOWN TO GLASS BARRIER AND WAIT 24hrs TO FULLY BOND.
8. CLEAN UP ANY EXCESS SILICONE.

NOTE: RAIL ENDS MUST BE ATTACHED TO STRUCTURE OR STRUCTURAL POST.
EXTRUSION JOINS MUST HAVE A SUITABLE JOINER PLATE

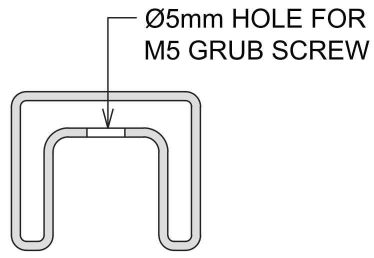
## REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS



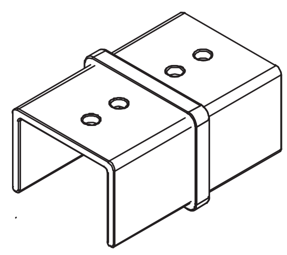

EPDM GASKET SYSTEM

TAPE & SILICONE SYSTEM

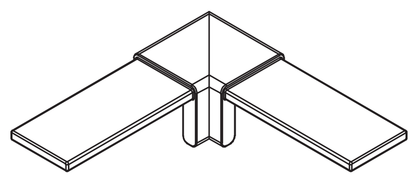
## COMPONENTS OVERVIEW



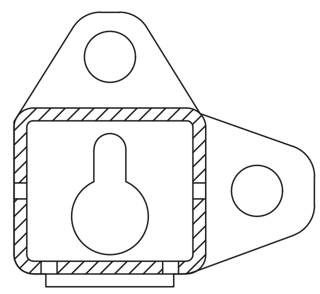

S25 STAINLESS RAIL



180° JOINER (& ADJUSTABLE VERTICAL JOINER)

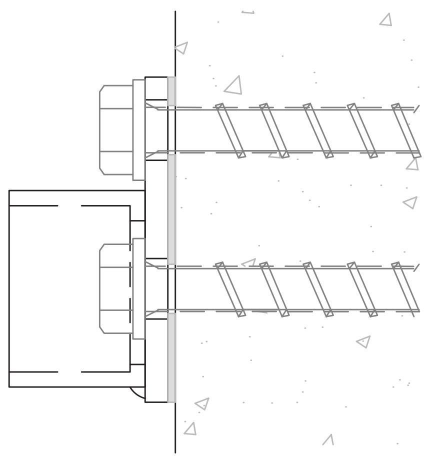


90° JOINER (& ADJUSTABLE HORIZONTAL JOINER)

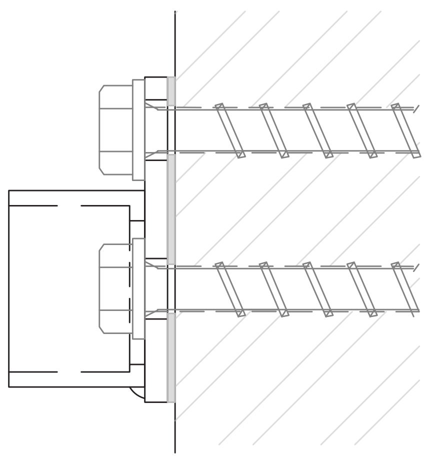


WALL BRACKET

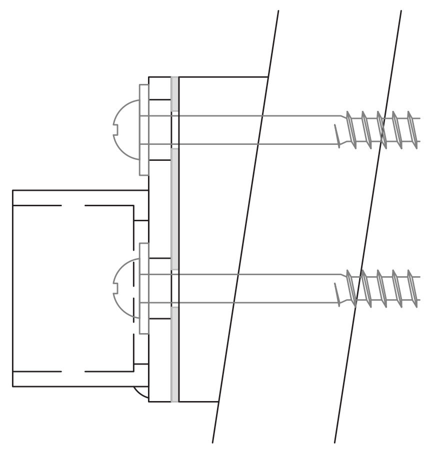
## INSTALL OVERVIEW WALL BRACKET



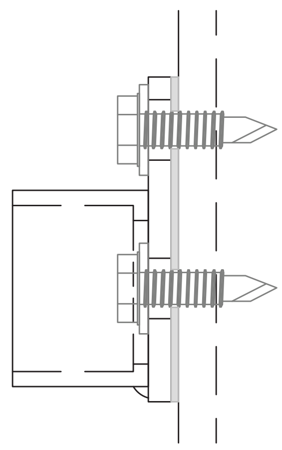
WALL BRACKET TO CONCRETE



WALL BRACKET TO BLOCKWALL



WALL BRACKET TO WEATHERBOARD

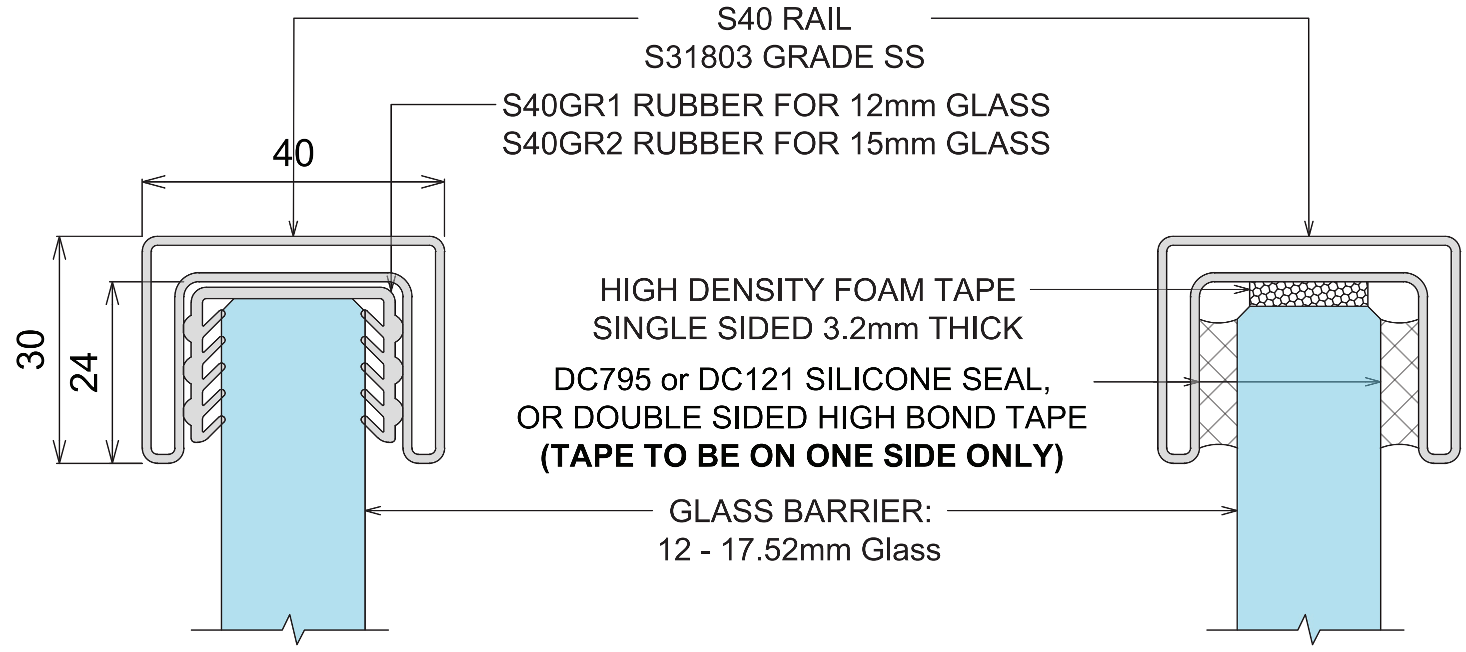


WALL BRACKET TO STEEL

**NOTES:**

1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
3. INTERLINKING RAIL END CONNECTION BRACKETS & ATTACHMENT DETAILS ARE SHOWN ON DRAWINGS WALL BRACKET INSTALL OPTIONS. ALL SCREWS TO BE STAINLESS STEEL WITH A MINIMUM ULTIMATE SHEAR STRENGTH OF 3.5kN (PER SCREW).
4. LINK RAIL SECTION AND CONNECTION PIECES TO BE S31803 GRADE STAINLESS STEEL, IN ACCORDANCE WITH NZS 4673:2001.
5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

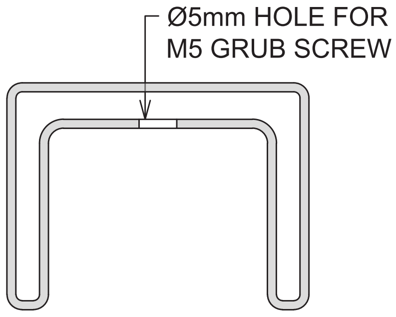
## REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS



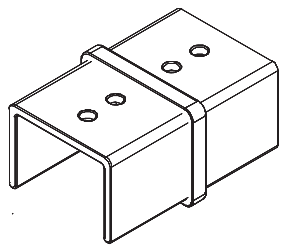

EPDM GASKET SYSTEM

TAPE & SILICONE SYSTEM

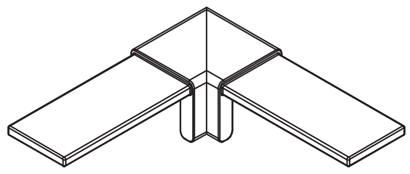
## COMPONENTS OVERVIEW



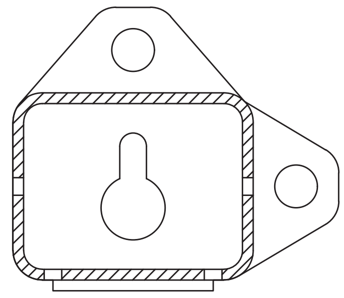

S40 STAINLESS RAIL



180° JOINER

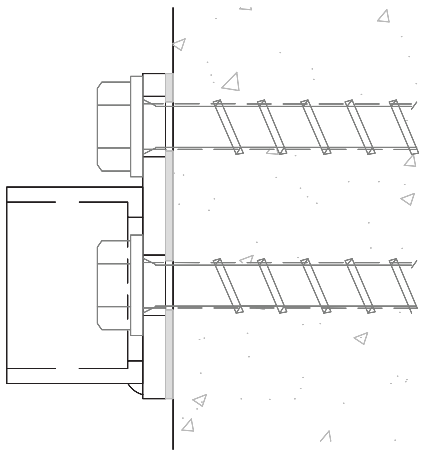


90° JOINER

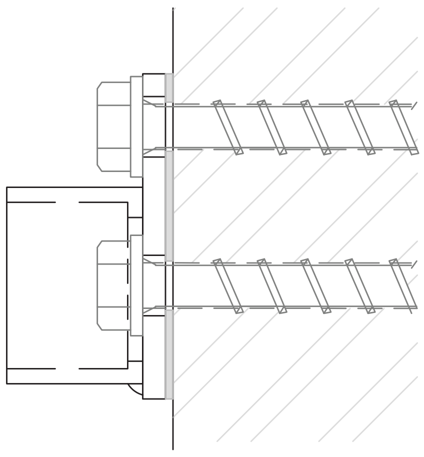


WALL BRACKET

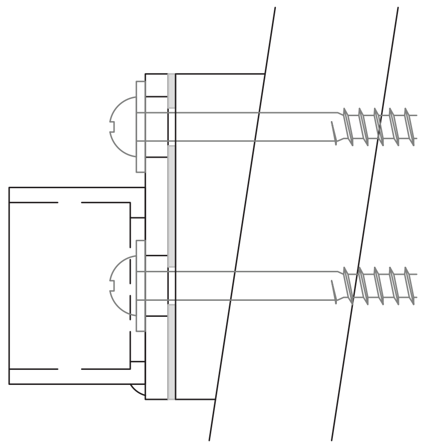
## INSTALL OVERVIEW WALL BRACKET



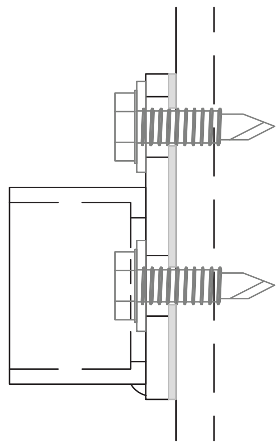
WALL BRACKET TO CONCRETE



WALL BRACKET TO BLOCKWALL



WALL BRACKET TO WEATHERBOARD



WALL BRACKET TO STEEL

**NOTES:**

1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
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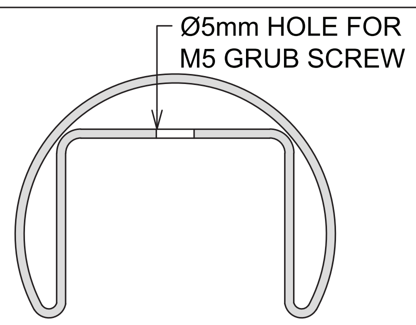
5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

## REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS


EPDM GASKET SYSTEM

TAPE & SILICONE SYSTEM

## COMPONENTS OVERVIEW



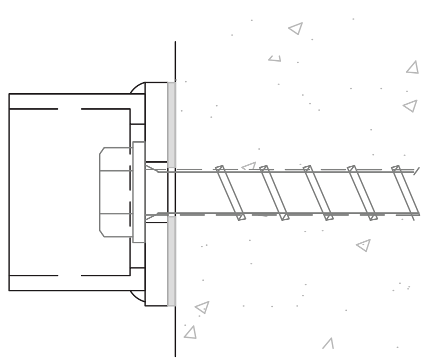

S40 STAINLESS RAIL

180° JOINER

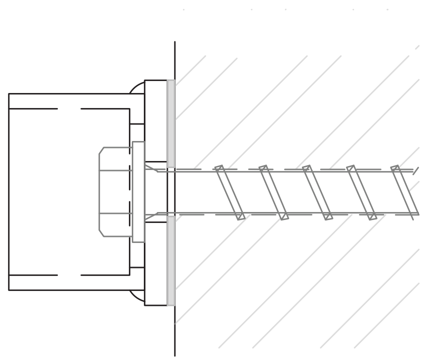
90° JOINER

WALL BRACKET

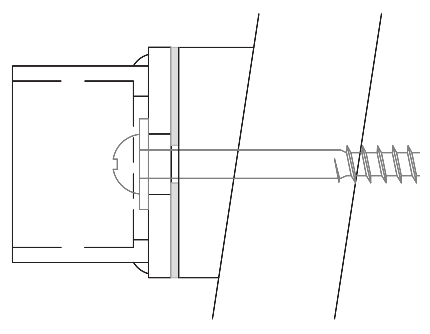
## INSTALL OVERVIEW WALL BRACKET



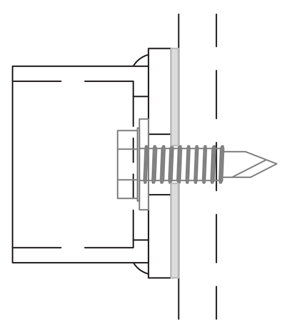
WALL BRACKET TO CONCRETE



WALL BRACKET TO BLOCKWALL



WALL BRACKET TO WEATHERBOARD



WALL BRACKET TO STEEL

**NOTES:**

1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
3. INTERLINKING RAIL END CONNECTION BRACKETS & ATTACHMENT DETAILS ARE SHOWN ON DRAWINGS WALL BRACKET INSTALL OPTIONS. ALL SCREWS TO BE STAINLESS STEEL WITH A MINIMUM ULTIMATE SHEAR STRENGTH OF 3.5kN (PER SCREW).
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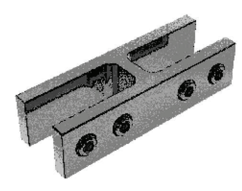
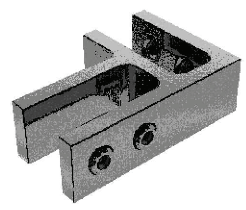
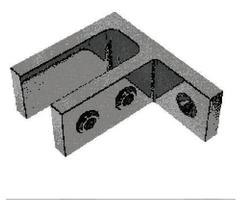
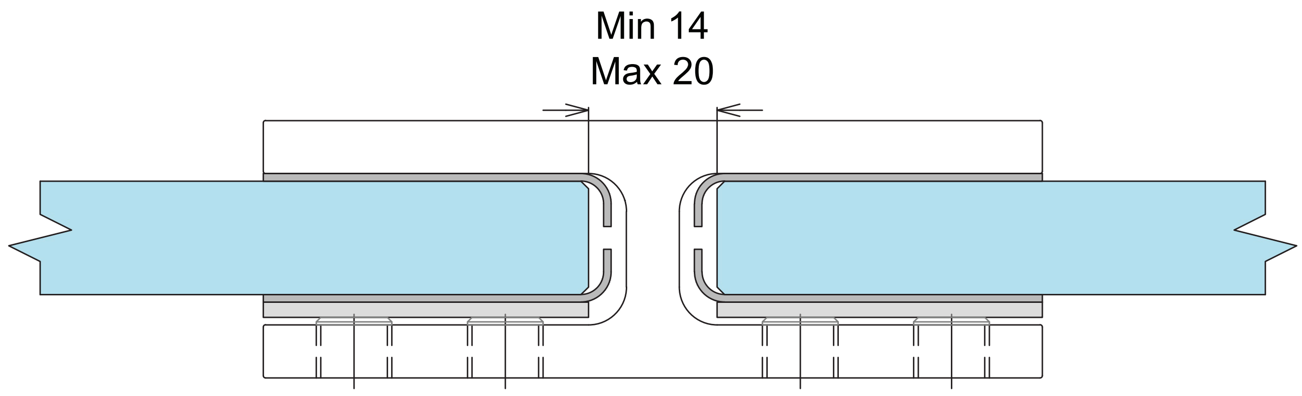
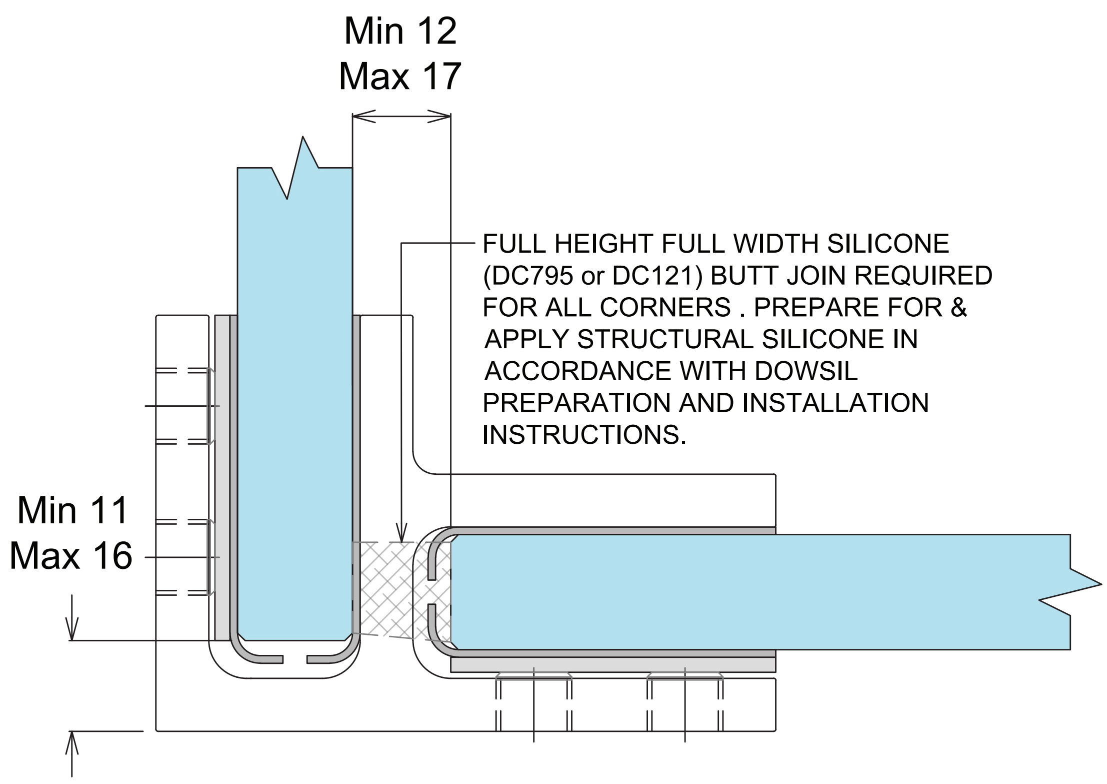
5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

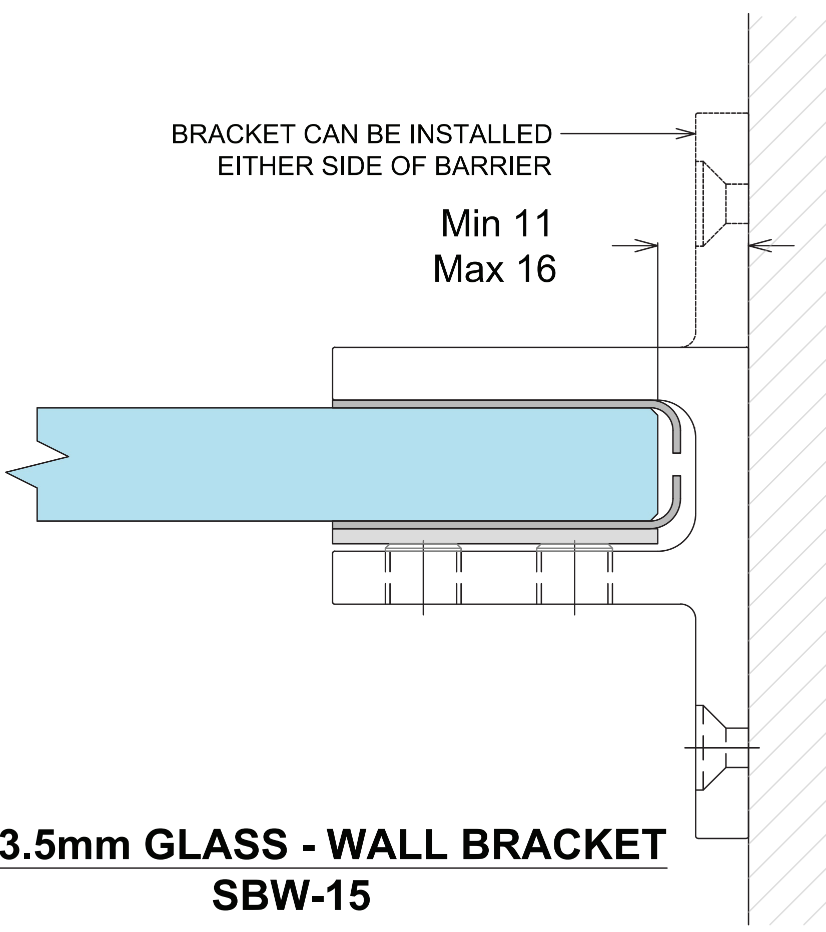
| Product | Code | Description |
|---|---|---|
|  | SB180-15<br>(180° bracket) | Suit glass from 12 - 13.5mm.<br>Stiffener bracket - No holes required.<br>180° glass to glass.<br>Duplex 2205. Stainless Steel |
|  | SB90-15<br>(90° bracket) | Suit glass from 12 - 13.5mm.<br>Stiffener bracket - No holes required.<br>90° glass to glass.<br>Duplex 2205. Stainless Steel |
|  | SBW-15<br>(wall bracket) | Suit glass from 12 - 13.5mm.<br>Stiffener bracket - No holes required.<br>90° glass to wall.<br>Duplex 2205. Stainless Steel |

All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.



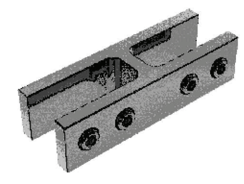
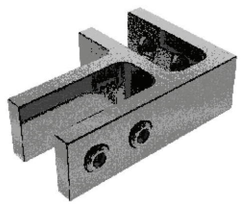
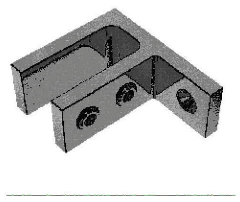

**12-13.5mm GLASS - STRAIGHT BRACKET**
**SB180-15**



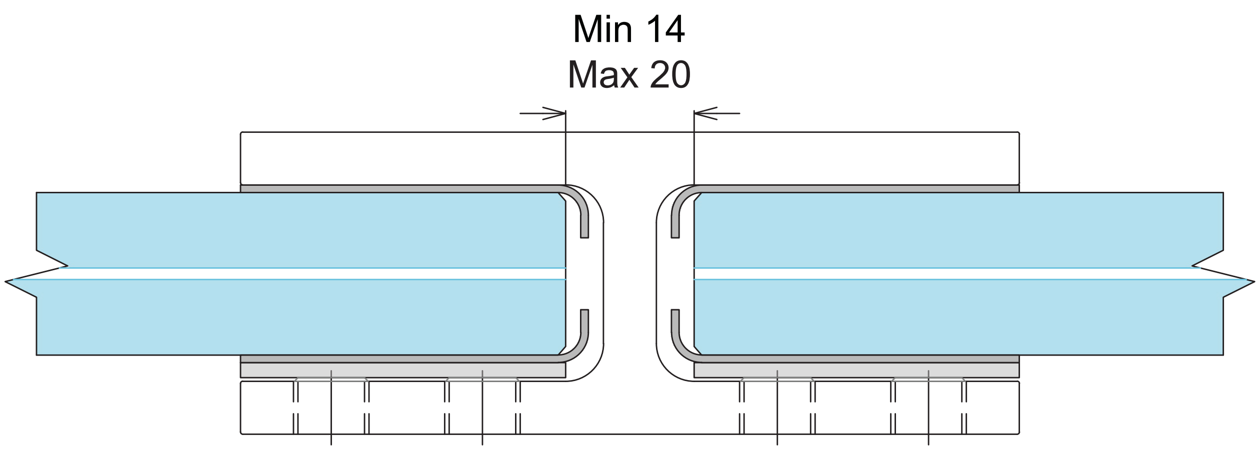




**12-13.5mm GLASS - CORNER BRACKET**
**SB90-15**

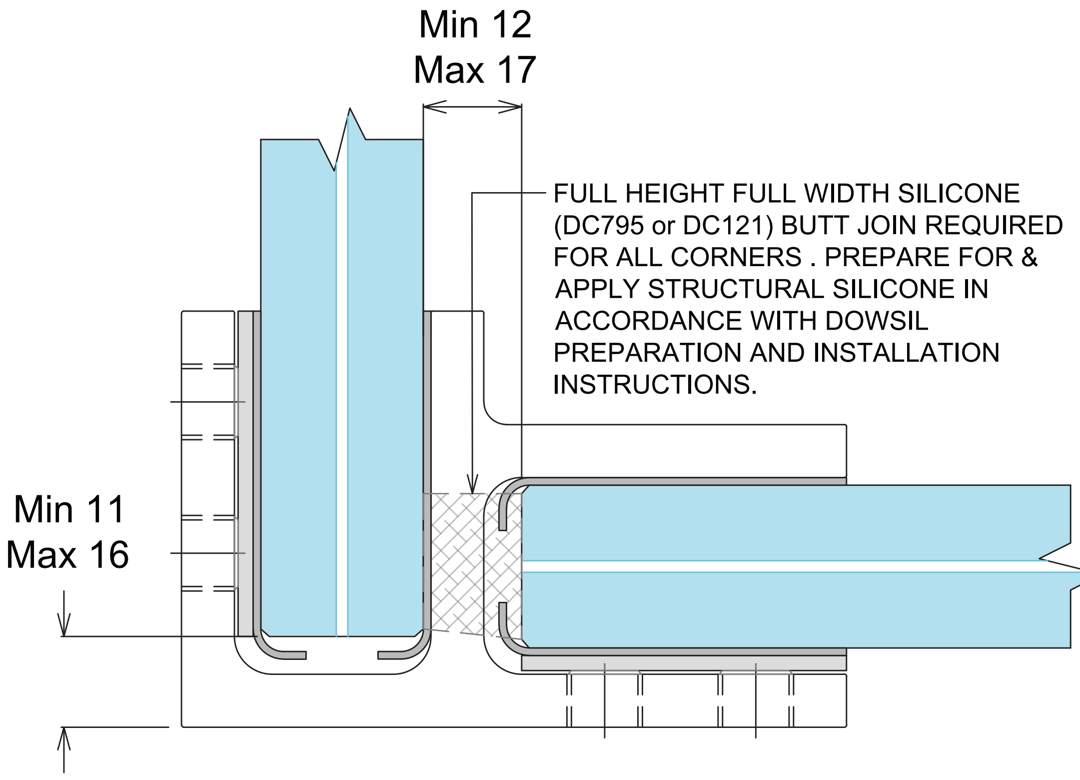
**12-13.5mm GLASS - WALL BRACKET**
**SBW-15**

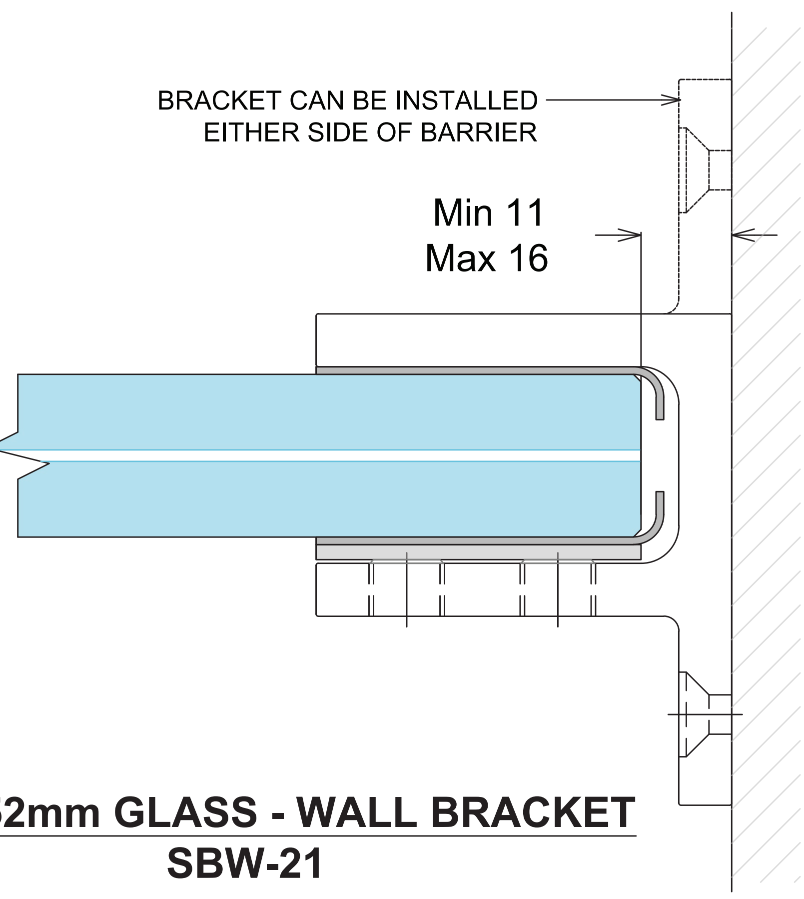
| Product | Code | Description |
|---|---|---|
|  | SB180-21<br>(180° bracket) | Suit glass from 17 - 17.52mm.<br>Stiffener bracket - No holes required.<br>180° glass to glass.<br>Duplex 2205. Stainless Steel |
|  | SB90-21<br>(90° bracket) | Suit glass from 17 - 17.52mm.<br>Stiffener bracket - No holes required.<br>90° glass to glass.<br>Duplex 2205. Stainless Steel |
|  | SBW-21<br>(wall bracket) | Suit glass from 17 - 17.52mm.<br>Stiffener bracket - No holes required.<br>90° glass to wall.<br>Duplex 2205. Stainless Steel |

All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.



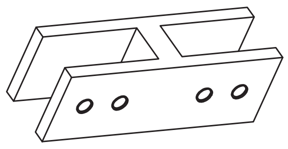
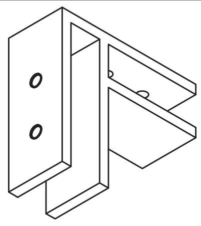
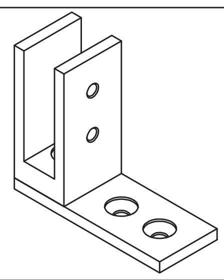

**17 - 17.52mm GLASS - STRAIGHT BRACKET**
**SB180-21**



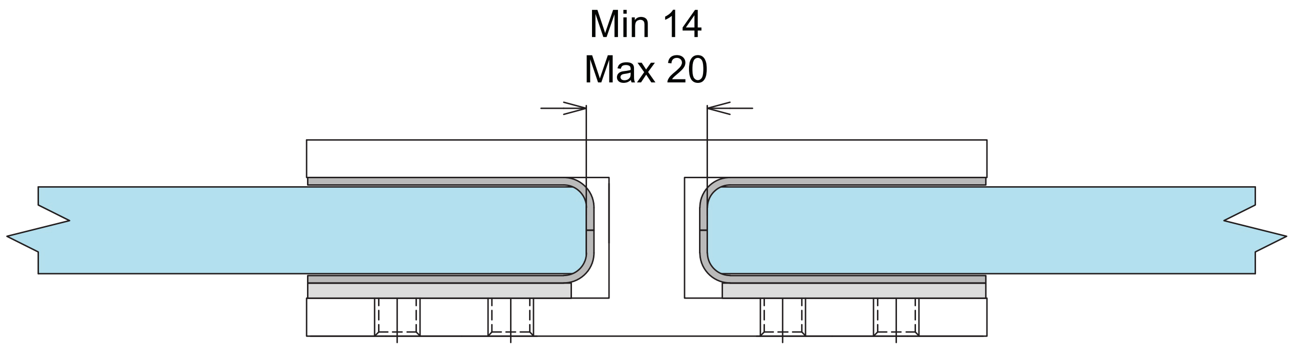




**17 - 17.52mm GLASS - CORNER BRACKET**
**SC90-21**

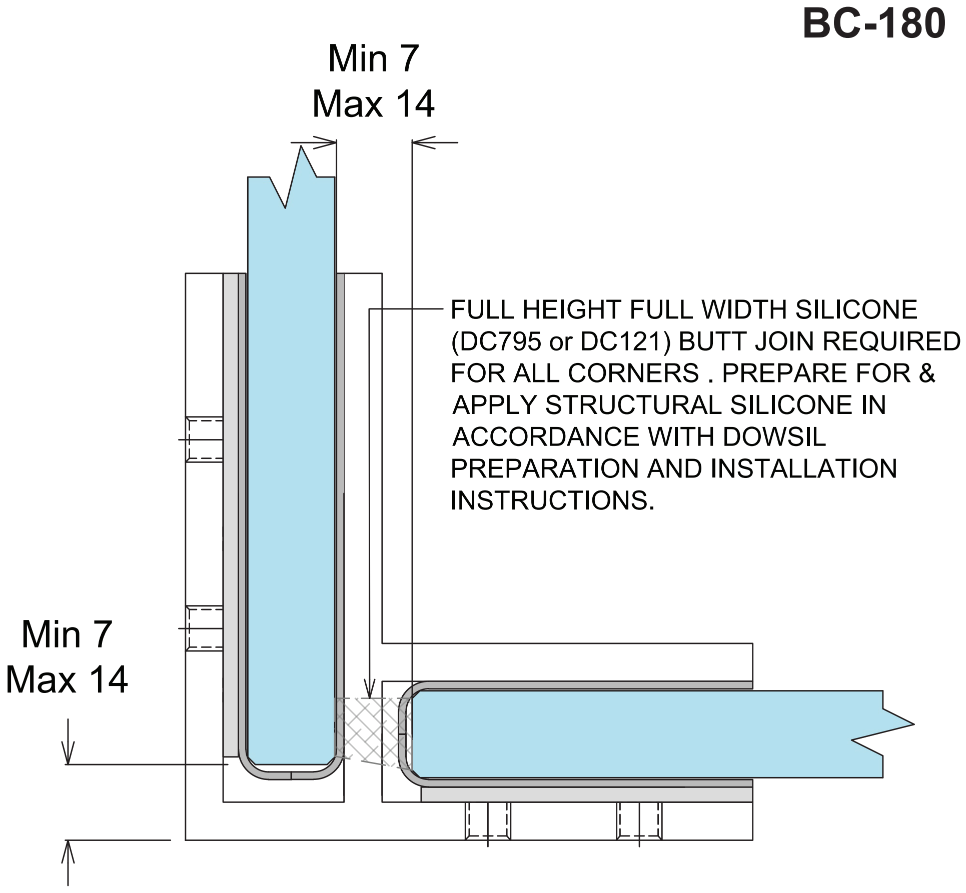
**17- 17.52mm GLASS - WALL BRACKET**
**SBW-21**

| Product | Code | Description |
| --- | --- | --- |
|  | BC180<br>(180° bracket) | Suit glass from 12 - 13.5mm.<br>180° glass to glass.<br>Duplex 2205. Stainless Steel<br>Available In Polished, Satin & Black |
|  | BCW<br>(90° bracket) | Suit glass from 12 - 13.5mm.<br>90° glass to glass.<br>Duplex 2205. Stainless Steel<br>Available In Polished, Satin & Black |
|  | BC-90<br>(wall bracket) | Suit glass from 12 - 13.5mm.<br>90° glass to wall.<br>Duplex 2205. Stainless Steel<br>Available In Polished, Satin & Black |

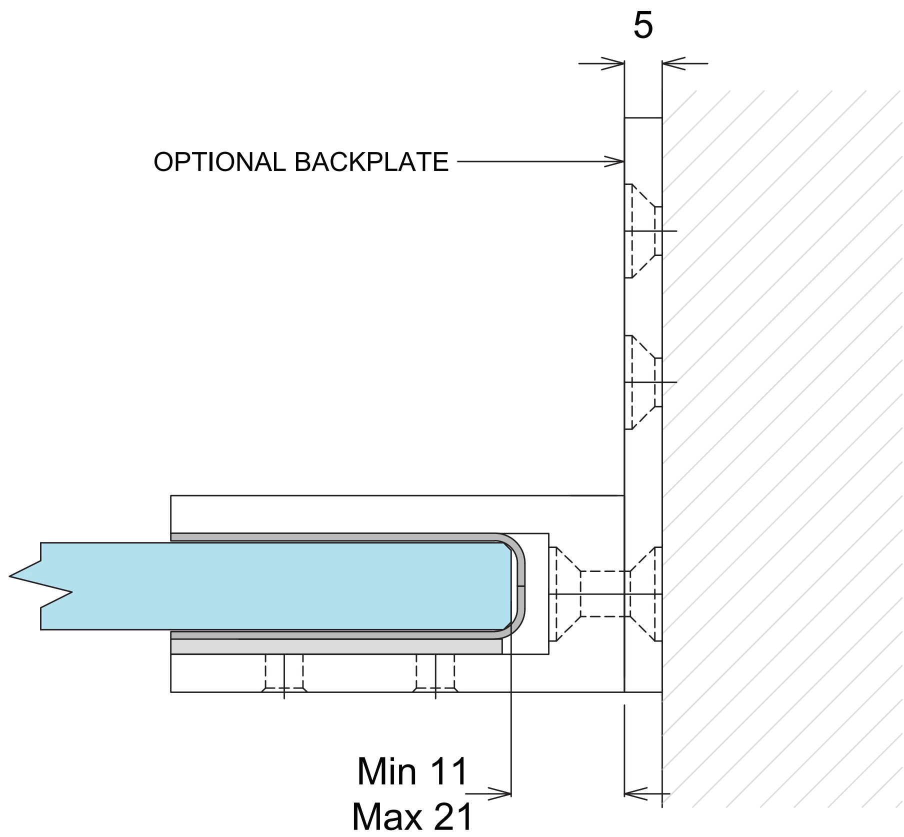
All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.




**12-13.5mm GLASS - STRAIGHT CLAMP**
**BC-180**

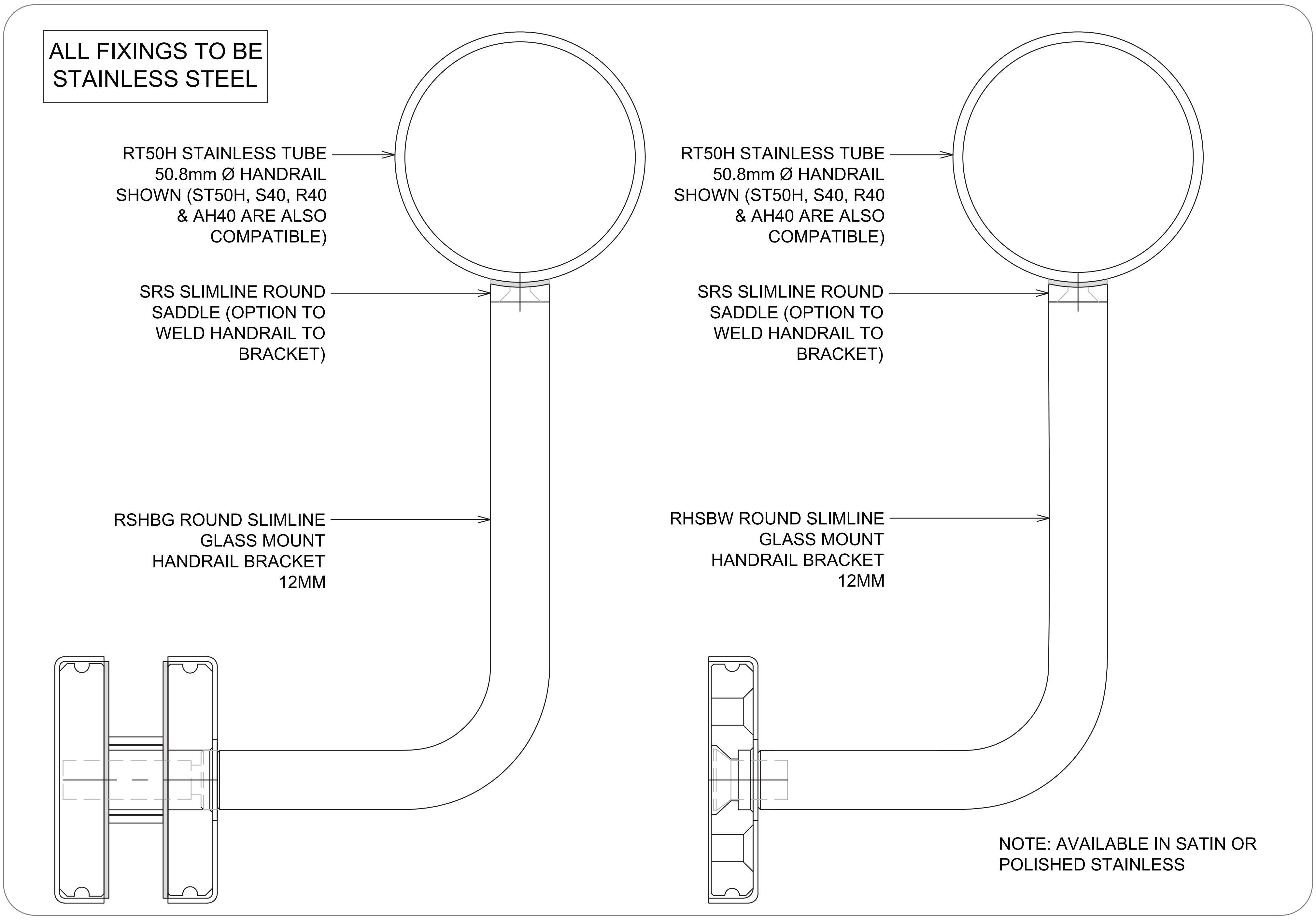


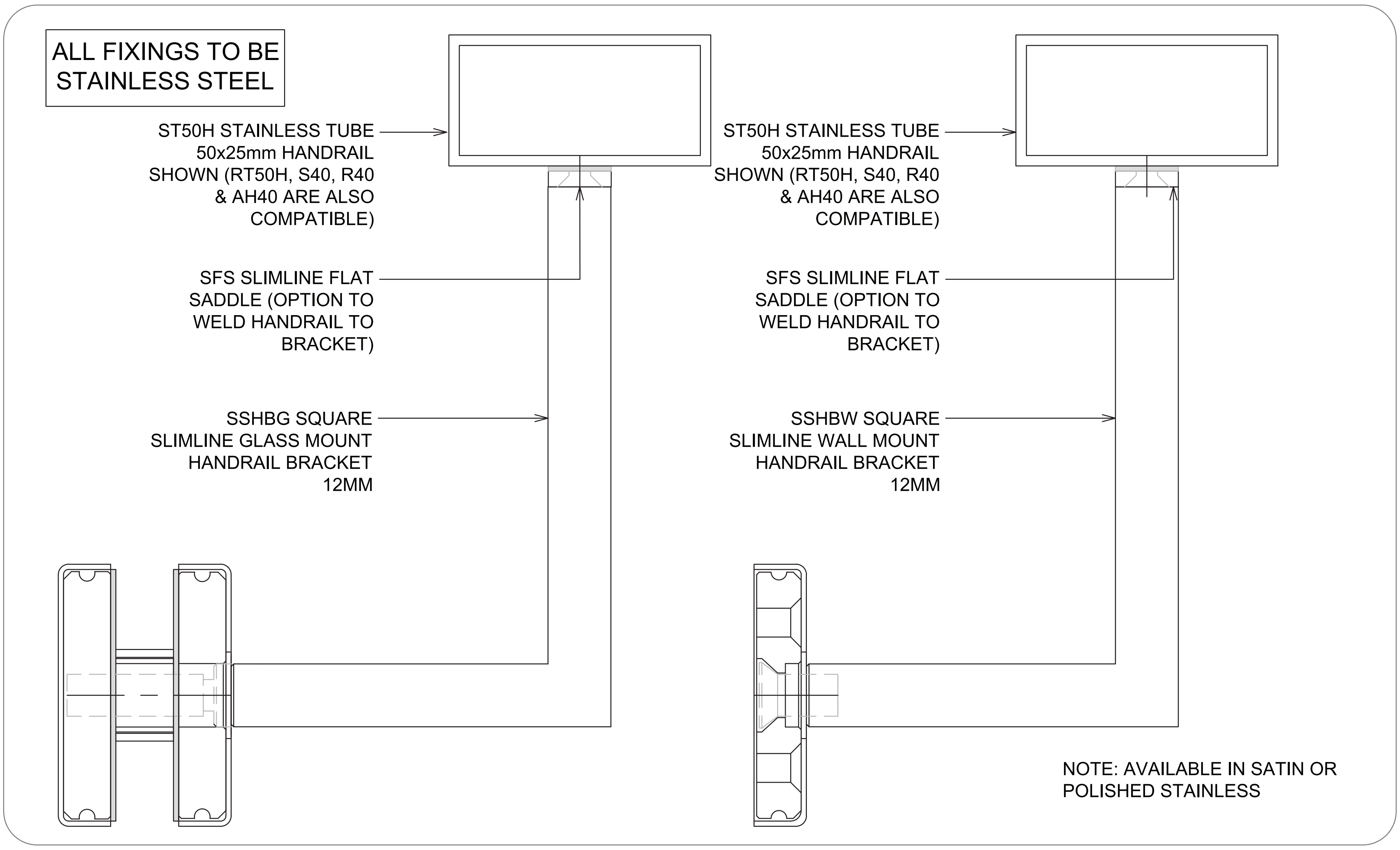

**12-13.5mm GLASS - CORNER BRACKET**
**BCW**



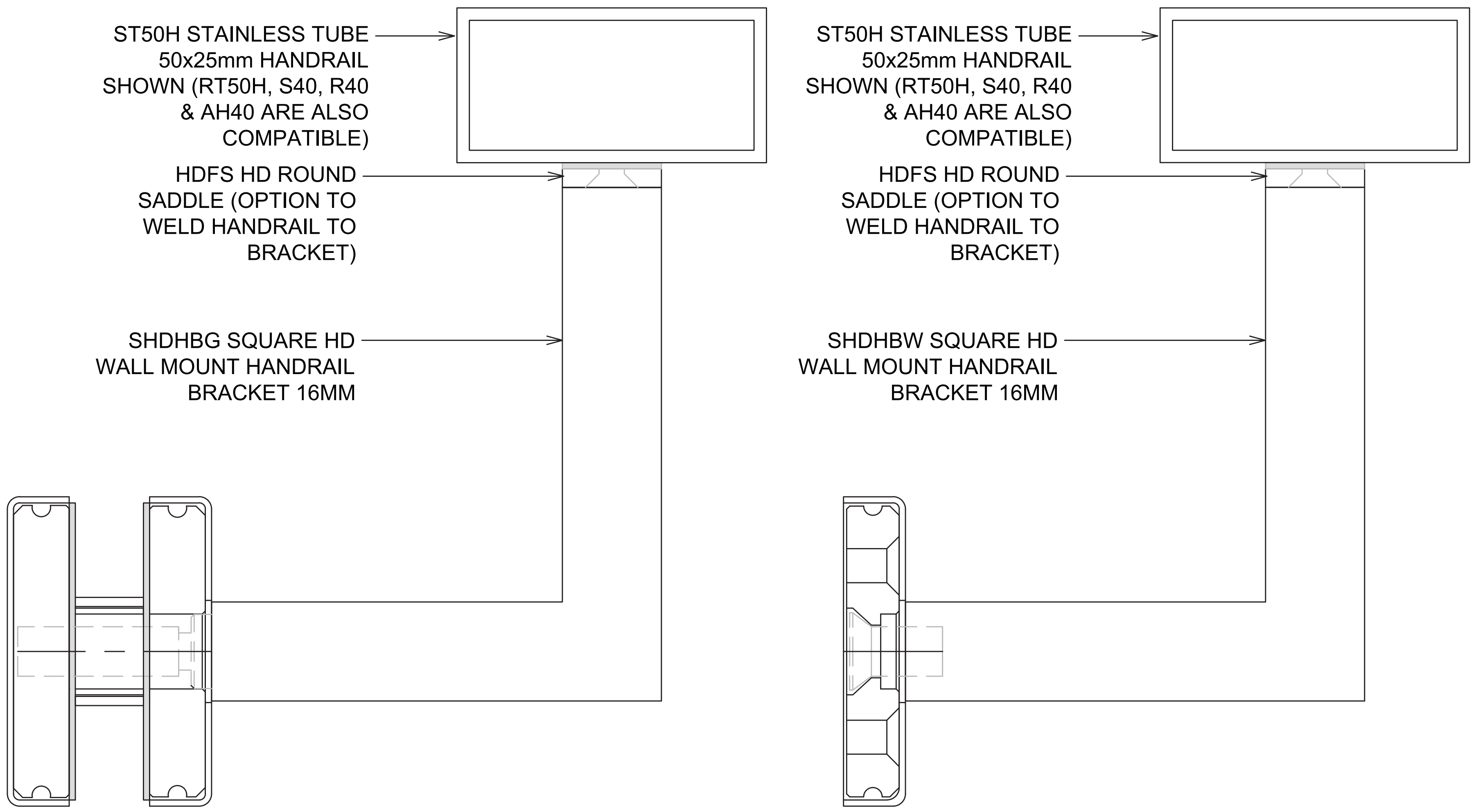

**12-13.5mm GLASS - WALL BRACKET**
**BC-90**

# VISTA BALUSTRADE SYSTEM
# SLIMLINE HANDRAIL BRACKETS

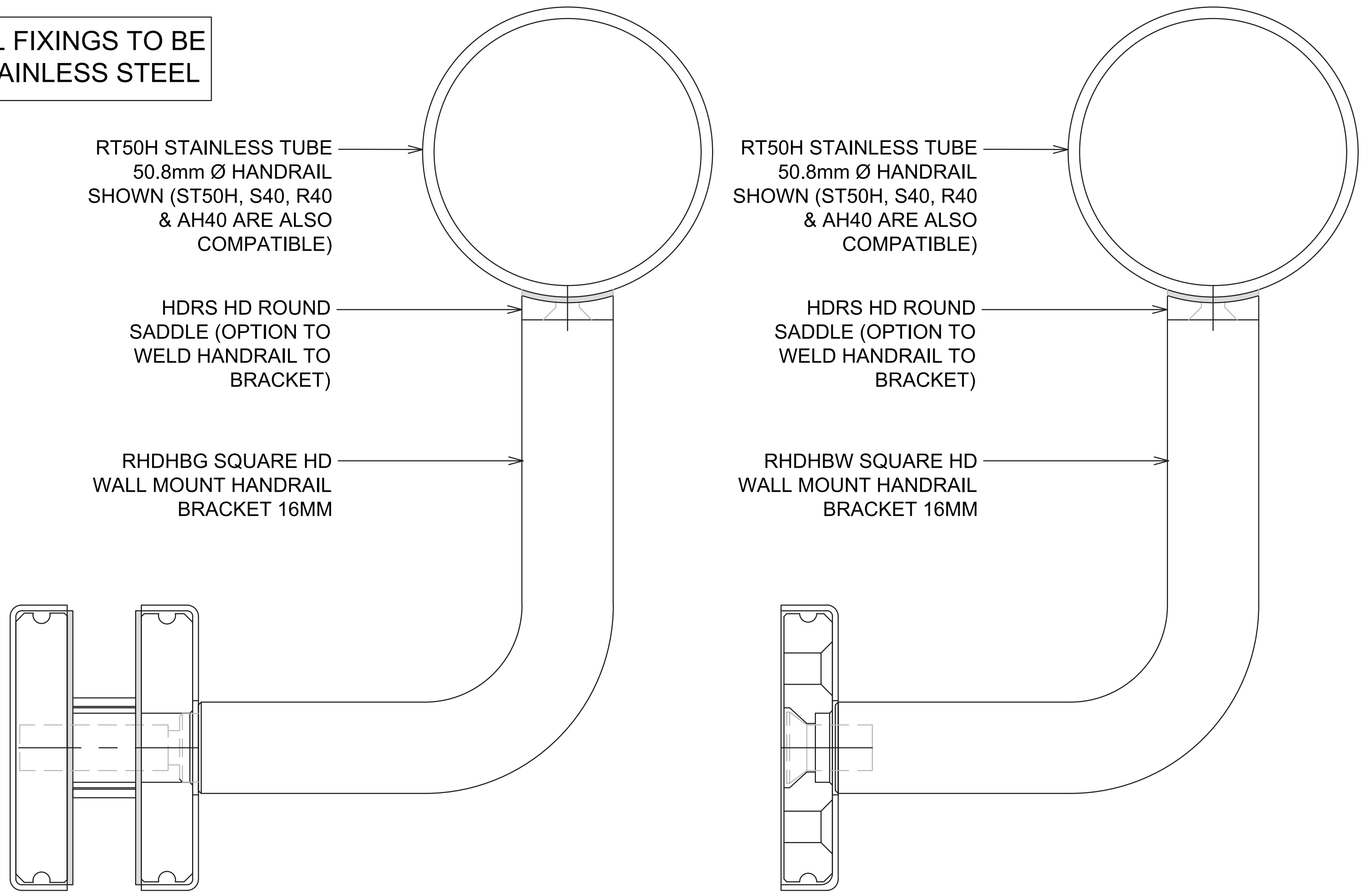


ALL FIXINGS TO BE STAINLESS STEEL




ALL FIXINGS TO BE STAINLESS STEEL



## CARE AND MAINTENANCE OF BALUSTRADE GLASS & HARDWARE

Hardware requires regular maintenance to ensure the system performs at its best. As a general rule, the harsher the environment, the more regular the maintenance required to keep your hardware in top condition. Also hardware or systems that are covered by verandas or wide eaves and not subject to natural rain wash needs regular cleaning to avoid damage to surface finish on both the aluminium and any surface coated hardware.

Fair wear and tear;

- **Seals and rubbers** will require replacing from time to time depending on the environment. As a general rule, they should last for 10 years or more, and can be replaced by service provider.
- **Tracks, rollers** (if accessible) and hardware require lubrication; rollers may require replacing due to normal wear and tear. This depends on the environment and amount of use.

## CARE AND MAINTENANCE, WASHING GLASS

Regular washing and drying of glass windows and doors are required to ensure their long term durability. In urban areas washing should be done every three to six months. The following guidelines apply:

a. When washing, soak the glass surface with warm water and a mild soap detergent solution or proprietary glass cleaners to loosen dirt and debris.

b. Use a soft grit free cloth or sponge when washing and try to avoid washing In direct sunlight. Do not use scrapers or razor blades.

c. After washing, rinse with clean water and then dry the glass using a clean, grit-free squeegee, cloth, or paper towel. Remember, wet glass is dirty glass.

d. All water and cleaning solution residue should be dried from the window gaskets, sealants, and frames to prevent water spots.

e. Avoid cleaning tinted and reflective glass surfaces in direct sunlight. When washing special glass, the following guidelines apply:

f. When washing double glazing and laminated glass, use the same procedures as above but ensure no solvents come into contact with the edge laminate interlayer or unit sealant.

g. With reflective or Low E coated surfaces, exercise special care when cleaning - special cleaners may be required as they can be hard to clean. Follow manufacturer's instructions.

h. It is advisable to check that frame drainage is not blocked this can affect laminate and insulated glass units.

DO NOT:

- Do Not Use Scrapers of any type or size on a Glass surface.
- Do Not - Leave building dirt or residues to remain on Glass for a period of time.
- Do Not - Begin cleaning glass until you have identified the surface type.
- Do Not - Clean Glass surfaces in direct sunlight.
- Do Not - Allow dirty water or cleaning residues to remain on the Glass.
- Do Not - Begin cleaning before rinsing off loose residues.
- Do Not - Use abrasive cleaning solutions, materials or solvents.
- Do Not - Allow metal parts of the cleaning equipment to come in contact with the Glass.
- Do Not - Trap abrasive particles between the cleaning material and the Glass.

DO:

- Clean glass promptly when dirt or building residues appear.
- Determine glass surface type.
- Exercise special care when cleaning coated surfaces.
- Avoid cleaning glass surfaces in direct sunlight.
- Start cleaning at the top of a building, then continue to lower levels.
- Soak the glass surface in a clean soapy solution before cleaning.
- Use a mild non abrasive commercial cleaner.
- Use a squeege to remove all cleaning solution.
- Try your procedures on a small window and check.

## CARE AND MAINTENANCE OF POWDERCOATING

Powder coating is available in a wide range of colours with commercially available surface integrity warranties from 10 to 30 years. The powder coating surface warranty period is conditional upon the formulation and micron thickness. Over time with exposure to the elements, powder coatings may show signs of weathering such as loss of gloss, chalking and slight color change. A simple regular clean will minimise the effects of weathering and will remove dirt, grime and other build-up detrimental to all powder coatings.

The frequency of such cleaning will depend on many factors including the:

- Geographical location of the building.
- Environment surrounding the building e.g. marine, industrial, alkaline or acidic, etc.
- Levels of atmospheric pollution including salts.
- Prevailing winds and the possibility of air borne debris causing erosive wear of the coating e.g. sand causing abrasion.
- Protection of part or all of the building by other buildings.
- Change in environmental circumstances during the lifetime of the building e.g. if rural became industrial.

The following guidelines apply:

a. Just a gentle clean with a soft brush and mild detergent, followed by a fresh water rinse, will maintain the long-term performance of the powder coated or anodised aluminium. In rural or normal urban environments cleaning should occur every six months. In areas of high pollution, such as industrial areas, geothermal areas or coastal environments, cleaning should occur every three months. In particularly hazardous locations, such as beachfronts, severe marine environments or areas of high industrial pollution, cleaning should be increased to monthly.
b. Sheltered areas can be at more risk of coating degradation than exposed areas. This is because wind-blown salt and other pollutants may adhere to the surface. These areas should be inspected and cleaned if necessary on a more regular basis.
c. Adequate on site protection of delivered and/or installed hardware must be provided. Hardware may get knocked, scratched, or splattered with mortar, plaster, or paint during the later stages of construction. If splashes occur immediately wash down the hardware unit affected with water or methylated spirits* (*wash area thoroughly afterwards). Do not allow splashes to harden.
d. To restore powder coated surfaces that have lost gloss or are chalking, polishing with a high quality crème polish in accordance with the manufacturer's instructions is recommended. Avoid polishes that contain cutting compounds, unless the surface is extremely weathered.

**DO NOT USE SOLVENTS** Strong solvent type cleaners should not be used. These are harmful to the extended life of your hardware system.

## CARE AND MAINTENANCE OF ANODISING

Anodised hardware is not only attractive, but also offers a durable and tough wearing finish. Some deterioration of the anodic oxide coating may occur, mainly as a result of grime deposition and subsequent attack by moisture, particularly if the moisture is contaminated with sulphur compounds.

Regular cleaning is essential to preserve the finish of anodised aluminium over a long period. The following guidelines apply:

e. Anodised aluminium should be washed with warm water and a suitable wetting agent or mild soap solution, in a similar manner to washing a car. A fine brush may be used to loosen dirt or grime. The use of anything stiffer or more abrasive may result in damage to the surface. Acid or alkali cleaners should not be used, as these will damage anodic films and may discolour coloured hardware.
f. Where greasy deposits or hard to remove grime is present, the anodising may be cleaned with a soft cloth dipped in white spirit, turpentine, kerosene, or a mild liquid scourer, followed by wiping it with a dry rag. However, the cleaner must ensure none of these solvents come into contact with other parts of the system. All solvents must be kept from contact with the Santoprene glazing gasket materials (the “rubber” seal around the glass), as most solvents will damage them.
g. It is essential to rinse anodised aluminium thoroughly with copious applications of clean water after cleaning, particularly where crevices are present, and then dry the glass to prevent water spots.

Regularly washing anodised hardware will ensure a long lasting product. In general, the following programme is recommended:

- Rural environments: every six months.
- Urban environments: every three months.
- Industrial and marine environments: every six months, as well as a monthly cold water wash.

For additional protection, especially in harsh environments, waxing with a good quality car wax after washing will assist in lifting and maintaining the appearance of your anodised hardware.

Damage to anodised surfaces may occur during building. Painters may accidentally splash paint on newly installed hardware, marring their appearance. The cleaner must act quickly and remove such splashes with a soft cloth moistened with water. Using water based paints allows the cleaner to clean with water - using solvents may put your hardware at risk.

## CARE AND MAINTENANCE OF STAINLESS STEEL

Stainless steel is used for fittings and hardware for its strength, aesthetics, and its inherent high level of corrosion resistance. 2205 & 316 Grade Stainless Steel (also known as Marine Grade stainless) hardware is predominately used for external applications such as balustrades, pool fences, canopies and spider walls, but many of the entry door patch fittings and handles use 304 Grade. The design of our frameless glass systems and installation techniques are intended to make the systems as maintenance free as possible, but there are still maintenance procedures, associated with any exposed structure, that must not be overlooked. Stainless Steel is called so as it is less prone to staining - it is important to recognize that the material is not impervious to mild staining or even corrosion in some instances. The likelihood and severity of staining is a product of exposure to marine salts and other corrosive materials which can affect installations even 20km inbound from coastal areas. Pollen and other airborne matter can also contribute to corrosion so there no areas of New Zealand where the topic can be ignored. The smoother the surface is, the less prone to discoloration.
Stainless steel may be discoloured from corrosion;

- If used in areas where rain does not wash the fittings.
- If it is exposed to a more aggressive environment than that for which the particular grade of steel is intended, e.g. highly polluted air, salt solutions or residues of cleaning agents containing chlorine.
- If it has a rough surface that enables a corrosive substance to adhere to.
- If the fittings are not cleaned after installation and have oil/acid from hands.
- If the design of the steel component has crevices and narrow gaps.
- If the surface is contaminated or damaged by grinding swarf or other iron particles from tools used in the installation work.
- If fasteners of ordinary steel or dissimilar metals are used for securing the hardware, or if the hardware comes into direct contact with adjacent components made of plain carbon steel in wet or humid conditions (galvanic corrosion).

Light corrosion is often referred to as “Tea Staining” as it is a brown surface discolouration which, although not affecting the material structurally, is none the less unsightly. The likelihood of this occurring can be greatly reduced through the use of 2205 or 316 Grade, regular cleaning and the use of protective coatings.

The following guidelines apply;

a. Any stainless steel hardware should be cleaned after installation and before any glass is installed.
b. Basic cleaning can be carried out using simple soap solutions or mild detergents applied with warm water (or proprietary cleaners) and a clean non-abrasive cloth. The solution should be thoroughly rinsed off with cold water, and wiped dry with a clean absorbent cloth. Isopropyl alcohol can also be used to clean finger and hand marks.
c. Avoid bleach, as this can mark the metal surface, and avoid any abrasive applicators - especially a ferrous based cleaning pad, such as steel wool, as this can introduce contamination which reacts with the stainless steel and makes corrosion worse.
d. Protective coatings are also available that can be easily applied to the hardware which can increase maintenance periods. Information regarding these products is available from OPUS. Stainless steel hardware should be cleaned again during the final clean and if possible after any surrounding building work is complete.
e. If mild tea staining has already occurred, a plastic abrasive pad - “Scotchbrite” for example - can be used, again with warm water and a mild detergent / soap solution. When abrading however, it is important to only rub in one direction which is the same direction as any visible brushed finish. Take care to only rub the steel components, not the glass. A stainless steel rejuvenating paste can also be used and works well in combination with a scotchbrite pad. Information regarding these products is available from OPUS.
f. When installation is complete the frequency of a regular cleaning regime will vary according to the installation design and level of exposure, especially in regard to proximity to the sea. In general cleaning should take place 3 to 4 times a year. Protective coatings can prolong the maintenance interval.

In summary:

- Cleaning should be thorough and at a frequency to suit the environment
- Wash with warm water and mild detergent or soap solution or use proprietary stainless steel cleaners.
- Rinse thoroughly with clean cold water and dry.
- Do not use harsh abrasive cleaners and especially no wire wool or similar ferrous scourers.
- If mild corrosion is present, then a mild abrasive detergent or rejuvenating paste can be used with a warm mild solution and fully rinsed with clean cold water.
- Heavier levels of staining can be removed with a light plastic abrading pad (such as Scotchbrite) rubbing must only be in one direction with that being as per any visible surface finish. This is best done with a rejuvenating paste  Use protective coatings after cleaning

Be careful to ensure cleaners do not effect fibre or other fitting gasket materials